

Movelt2 Configuration

Not a required course: The provided system has already been configured.

Movelt2 Configuration

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Movelt2 is an advanced software framework for robot motion simulation, planning, and control.

1. Environment Setup

ROS2: Humble

If you are using a Jetson Nano development board, you will need to enter the commands from the tutorial into the accompanying virtual machine to run this function.

1.1. Install Movelt2

```
sudo apt install ros-humble-moveit*
```

Jetson Orin series boards have been tested and can directly run Movelt2 related cases. The overall smoothness is acceptable!

1.2. Create Folder

The virtual machine has been created.

Create MoveIt2 auto-generated configuration files:

```
mkdir ~/dofbot_pro_ws/src/dofbot_pro_moveit
```

2. MoveIt2 Configuration

MoveIt Setup Assistant is a graphical tool used to simplify and accelerate the creation and configuration of MoveIt2 configuration packages.

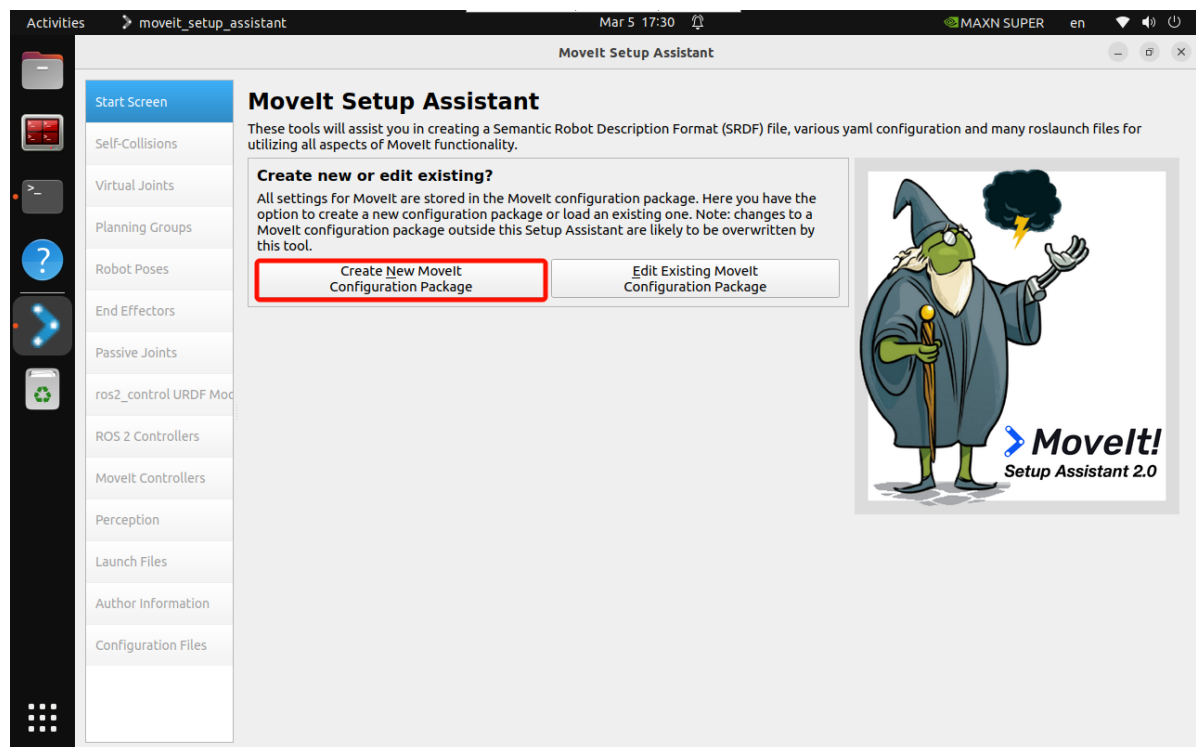
Through this tool, users can easily generate configuration files for robot motion planning and control without manually editing complex configuration files.

2.1. Start Assistant

Start MoveIt Setup Assistant:

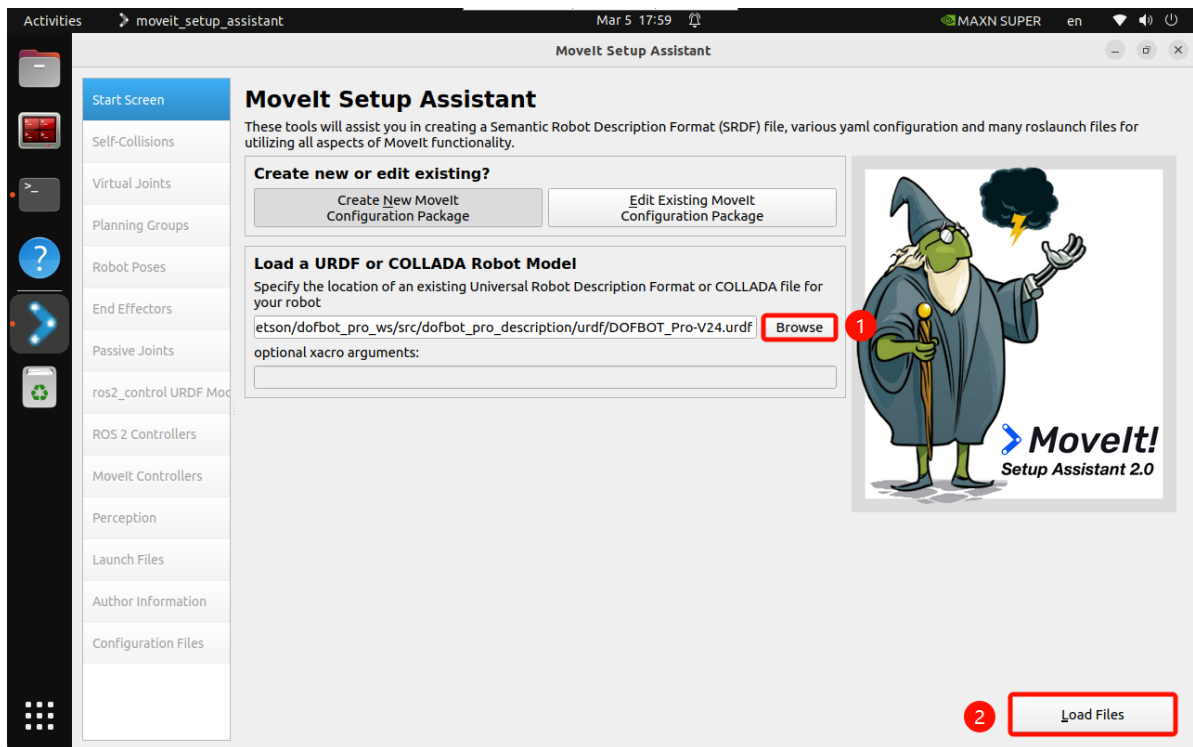
```
ros2 launch moveit_setup_assistant setup_assistant.launch.py
```

For first use, you can choose to create a new MoveIt configuration package. For later simple modifications, you can choose to edit an existing MoveIt package:

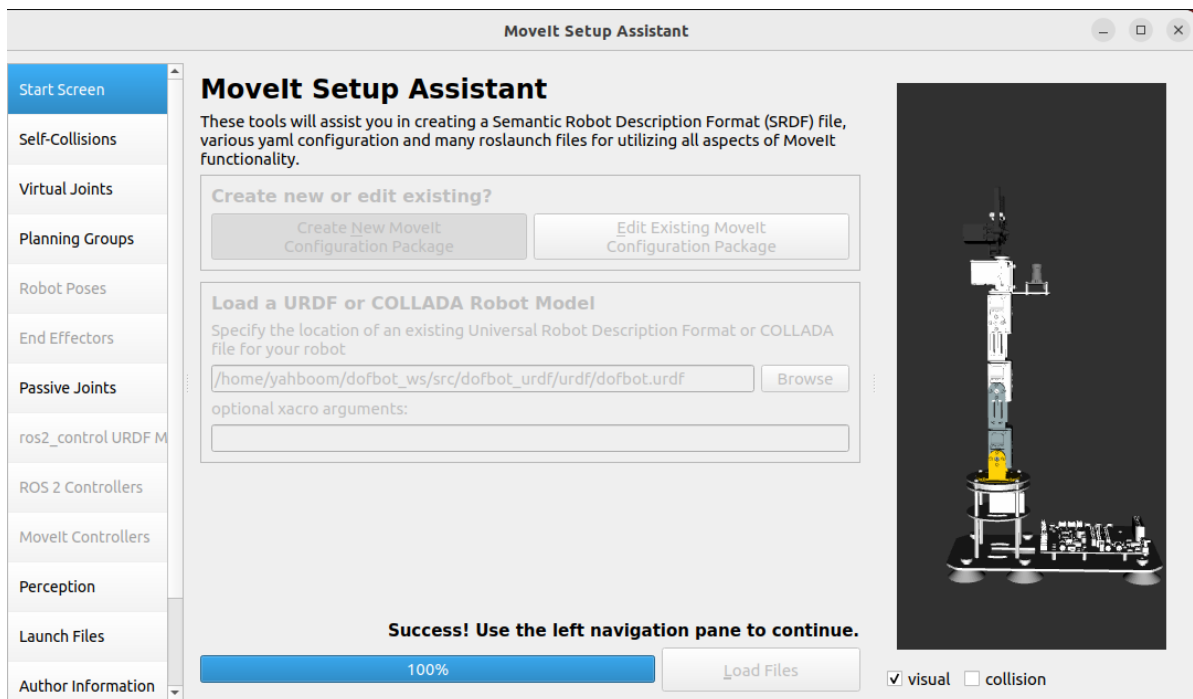


Creating a new MoveIt configuration package requires importing the robot's URDF model: Simply importing the URDF model file to MoveIt Assistant will cause an error. You need to use the compiled package (dofbot_pro_description) URDF file path

```
/home/yahboom/dofbot_ws/src/dofbot_urdf/urdf/dofbot.urdf
```



After selecting the URDF file, click Load Files, and MoveIt Assistant will display the robot model:

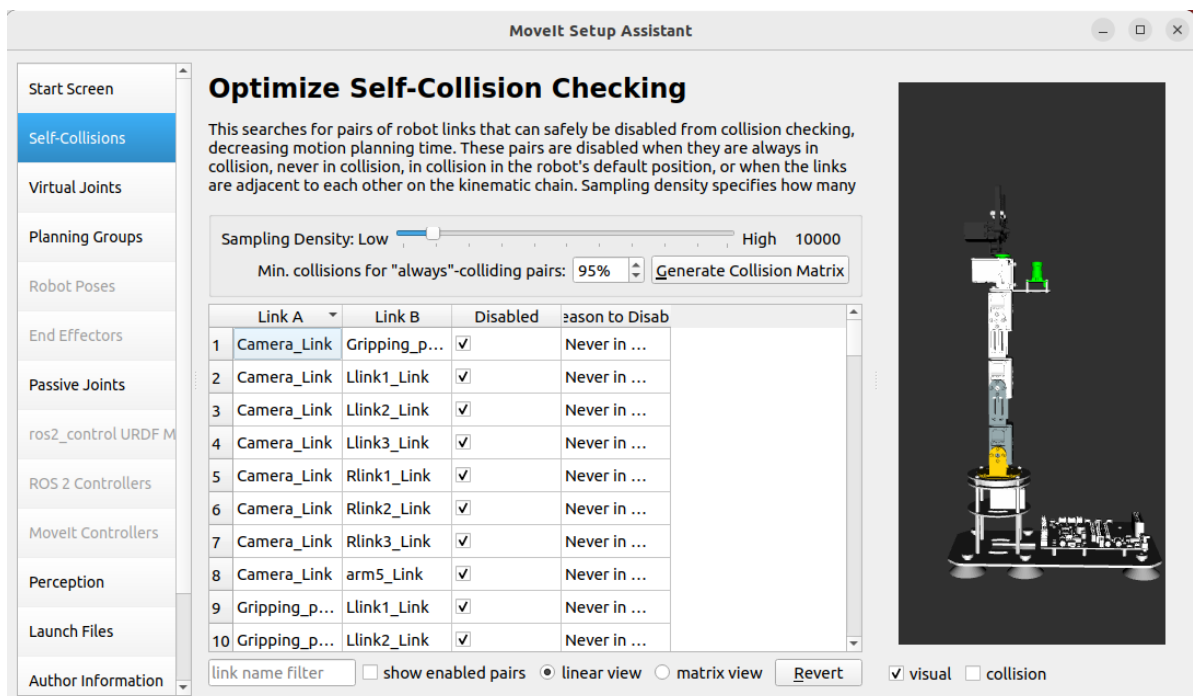
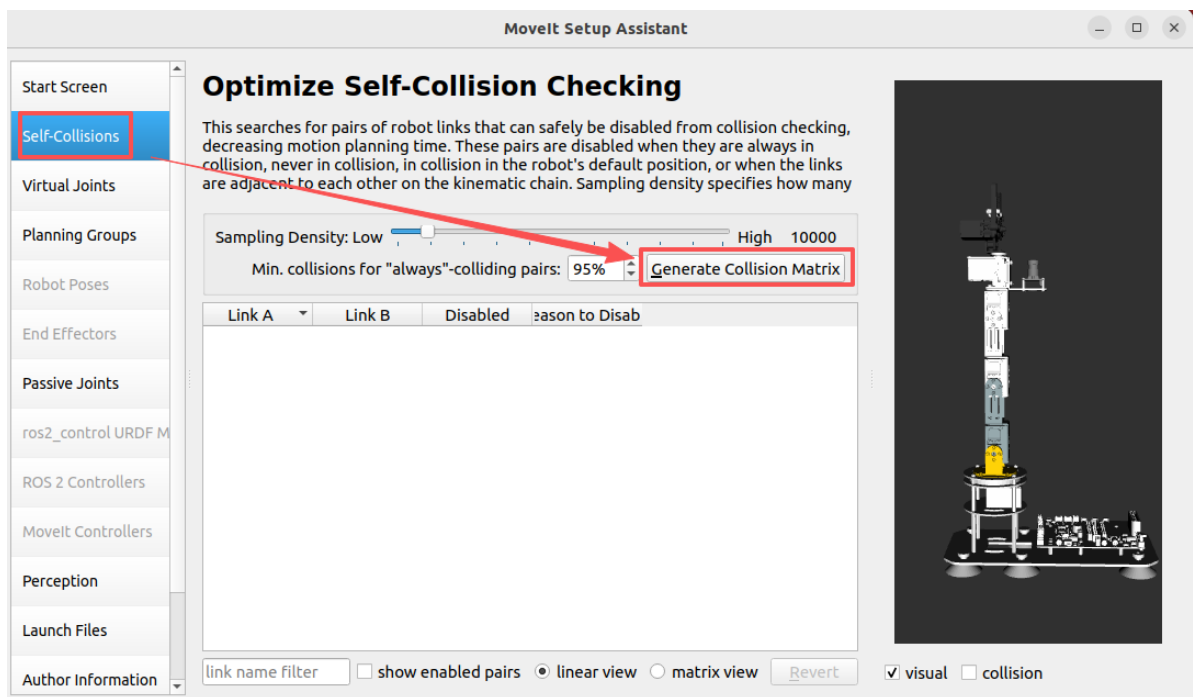


2.2. Collision Detection

The self-collision matrix is a feature used to optimize motion planning. Its main purpose is to generate a matrix that describes whether various links in the robot model may collide with each other.

This matrix can help the motion planner avoid unnecessary collision detection during path planning, thereby improving planning efficiency.

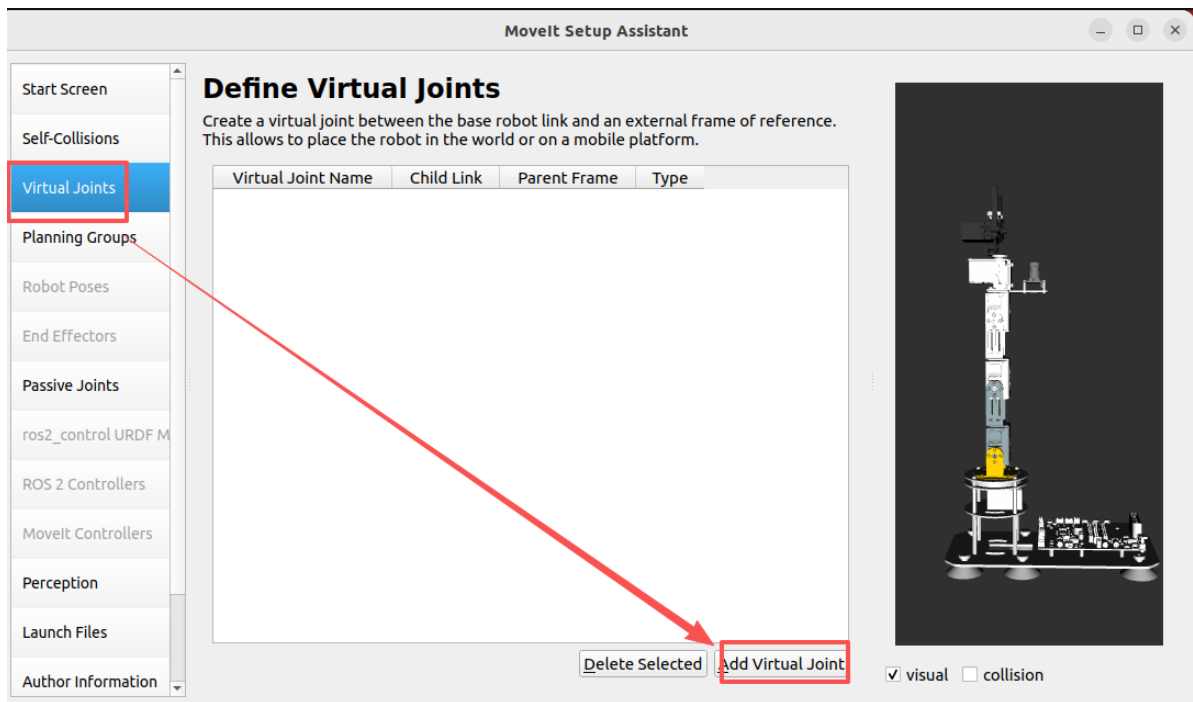
To generate the self-collision matrix, you can click the auto-generate option in MoveIt Setup Assistant:



2.3. Virtual Joints

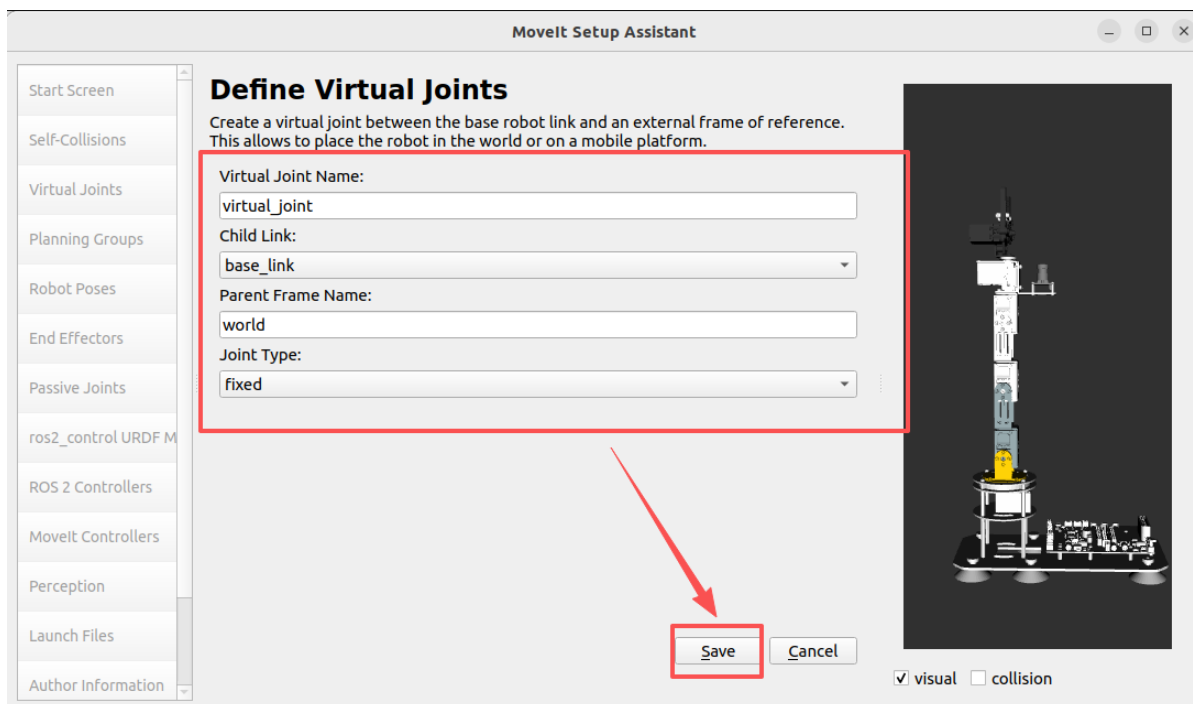
Virtual joints are used to define the relationship between the robot model and the external world.

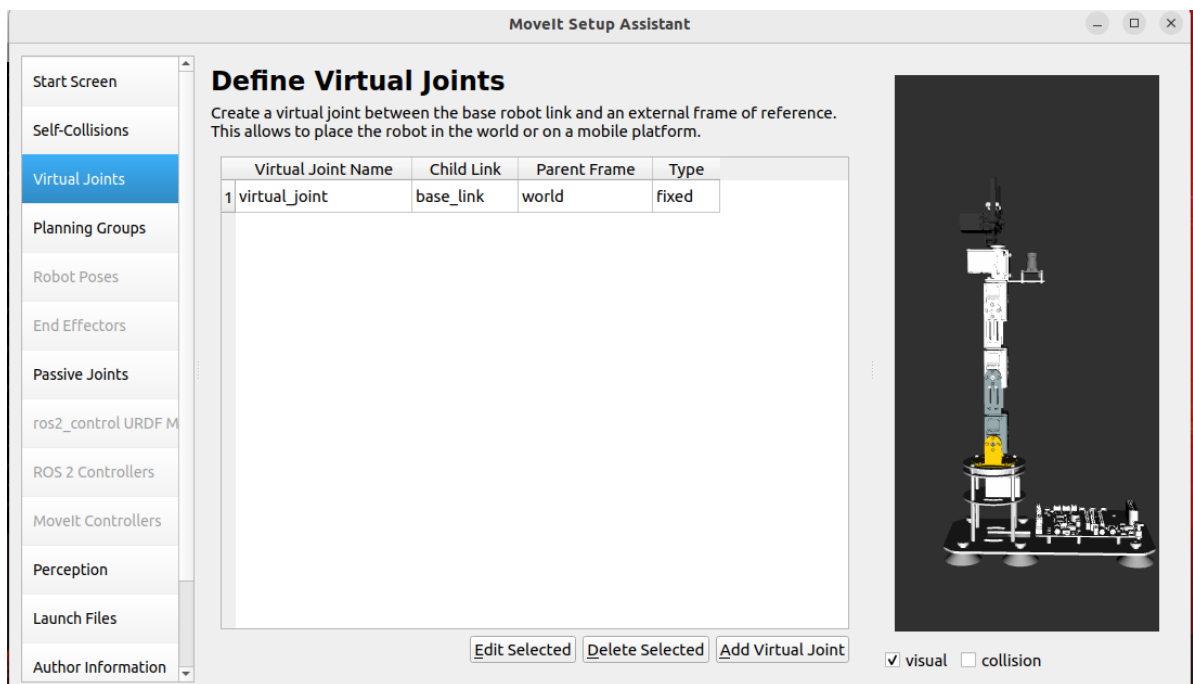
Virtual joints are typically used to describe the connection between the robot base and a fixed reference frame (such as the world coordinate system).



Fix the robot base link `base_link` and `world`:

- Virtual Joint Name: `virtual_joint`
- Child Link: `base_link`
- Parent Frame Name: `world`
- Joint Type: `fixed`

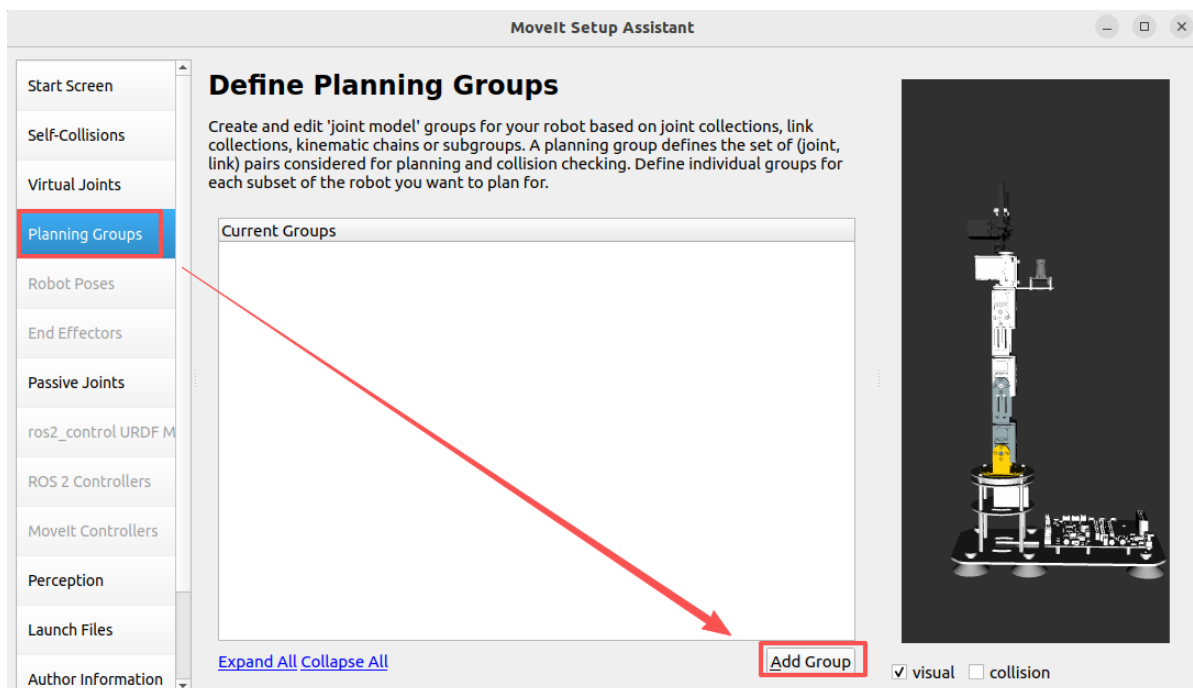




2.4. Motion Planning Groups

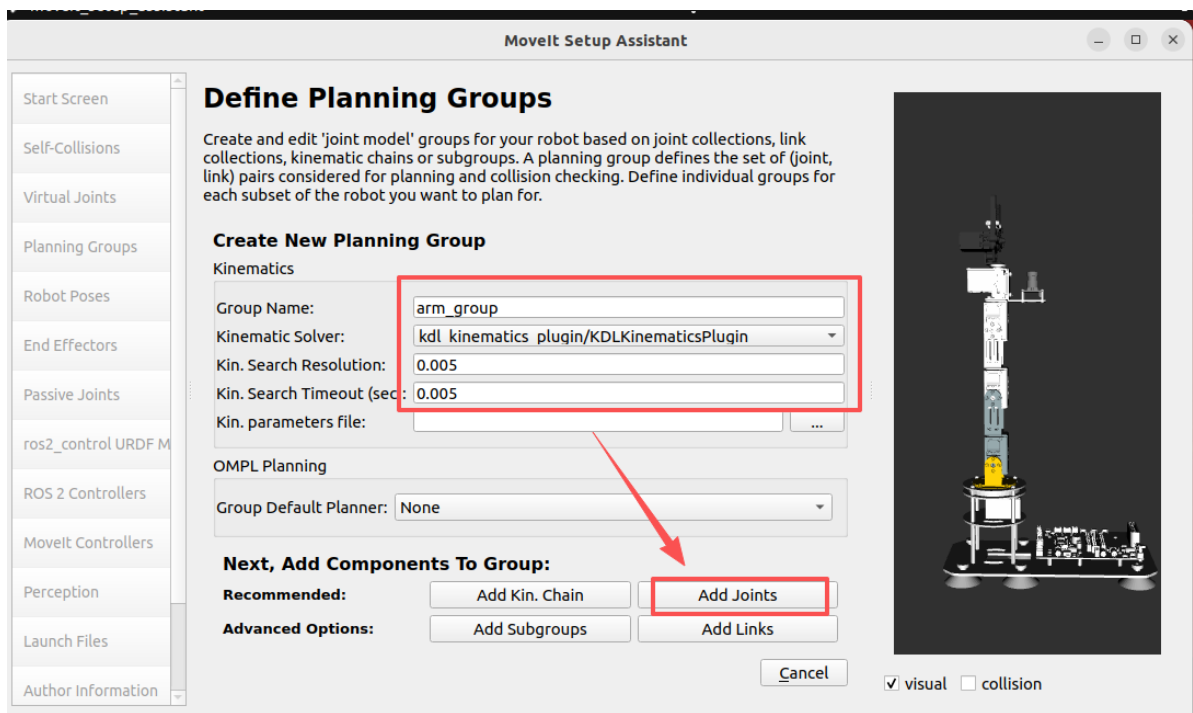
Planning groups are key steps in configuring robot motion planning.

Planning groups define which joints and links in the robot can move together and how their motion is planned.



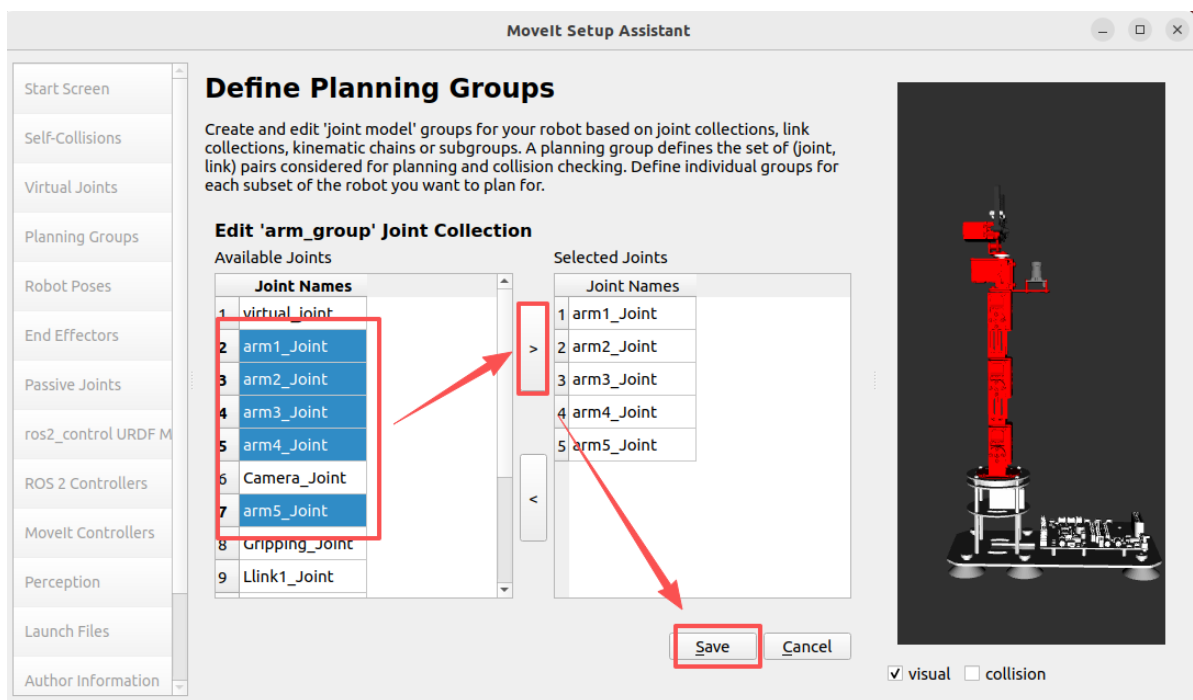
Plan robotic arm group: arm_group

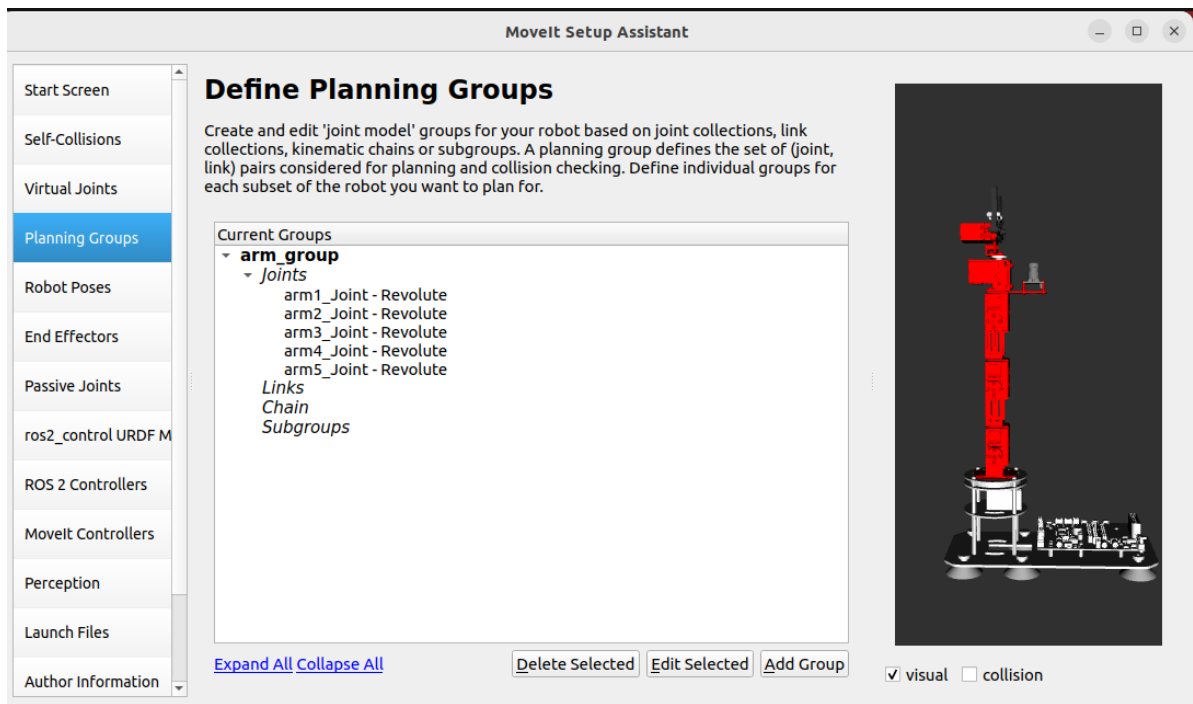
- Group Name: arm_group
- Kinematic Solver: kdl_kinematics_plugin/KDLKinematicsPlugin
- Kin. Search Resolution: 0.005
- Kin. Search Timeout: 0.005



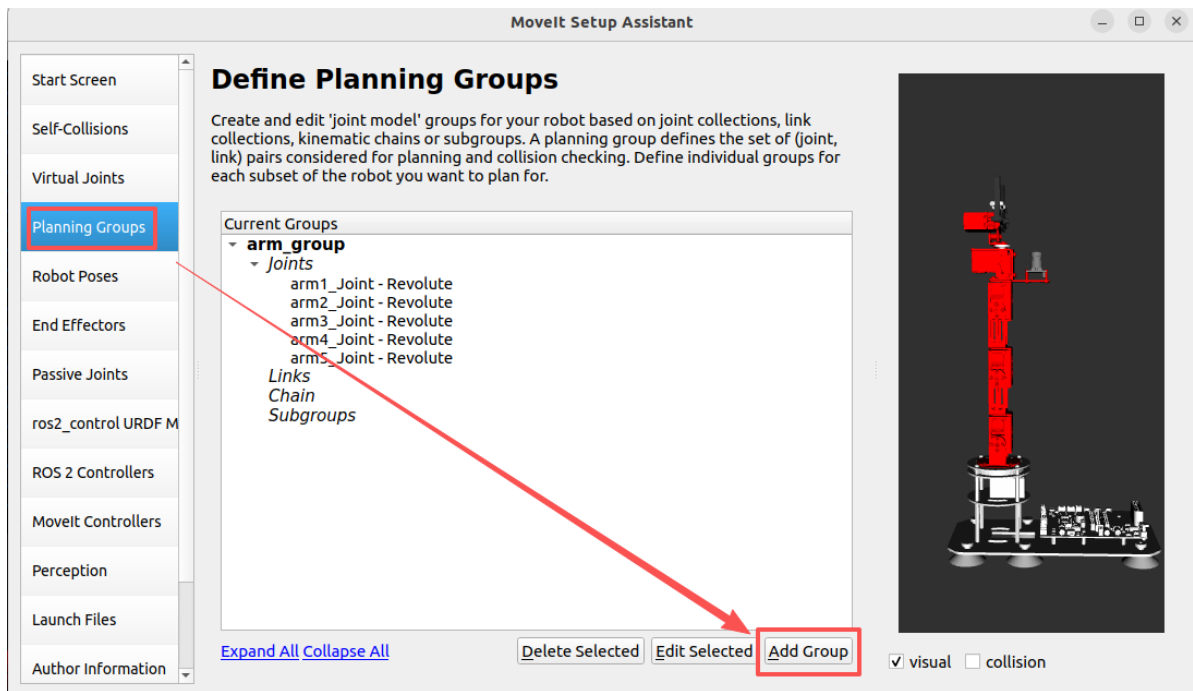
- Add Joints: Select the robotic arm joints

Joints: Arm1_Joint, Arm2_Joint, Arm3_Joint, Arm4_Joint, Arm5_Joint



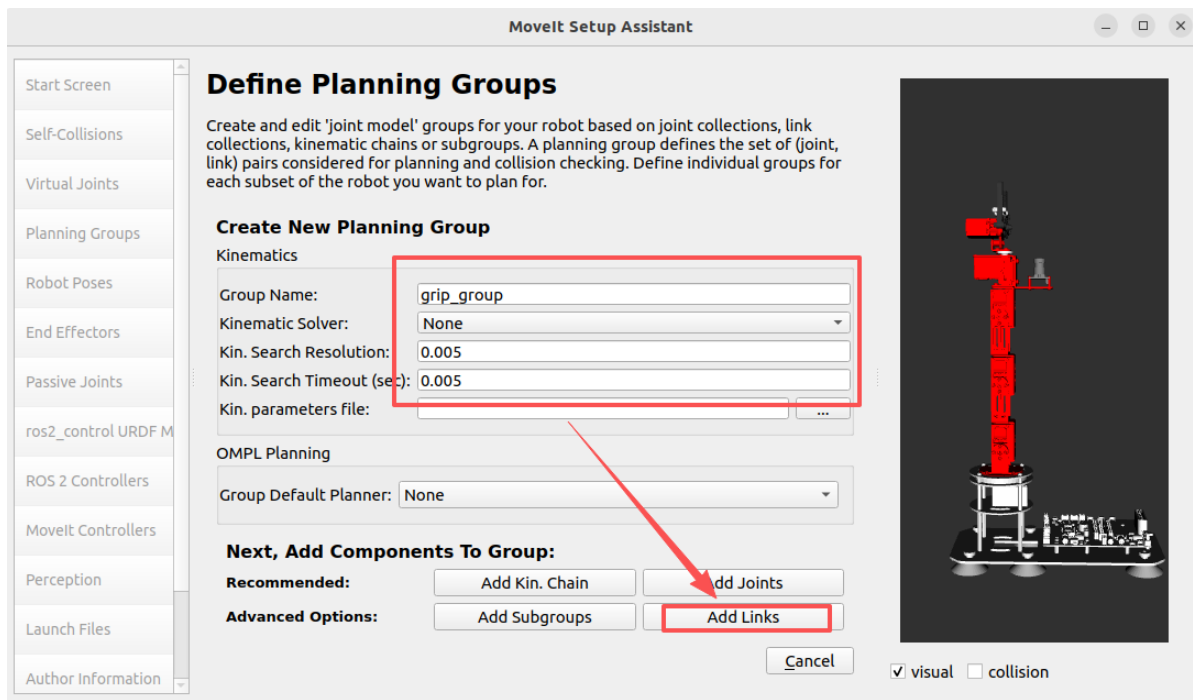


Plan Gripper Group:grip_group



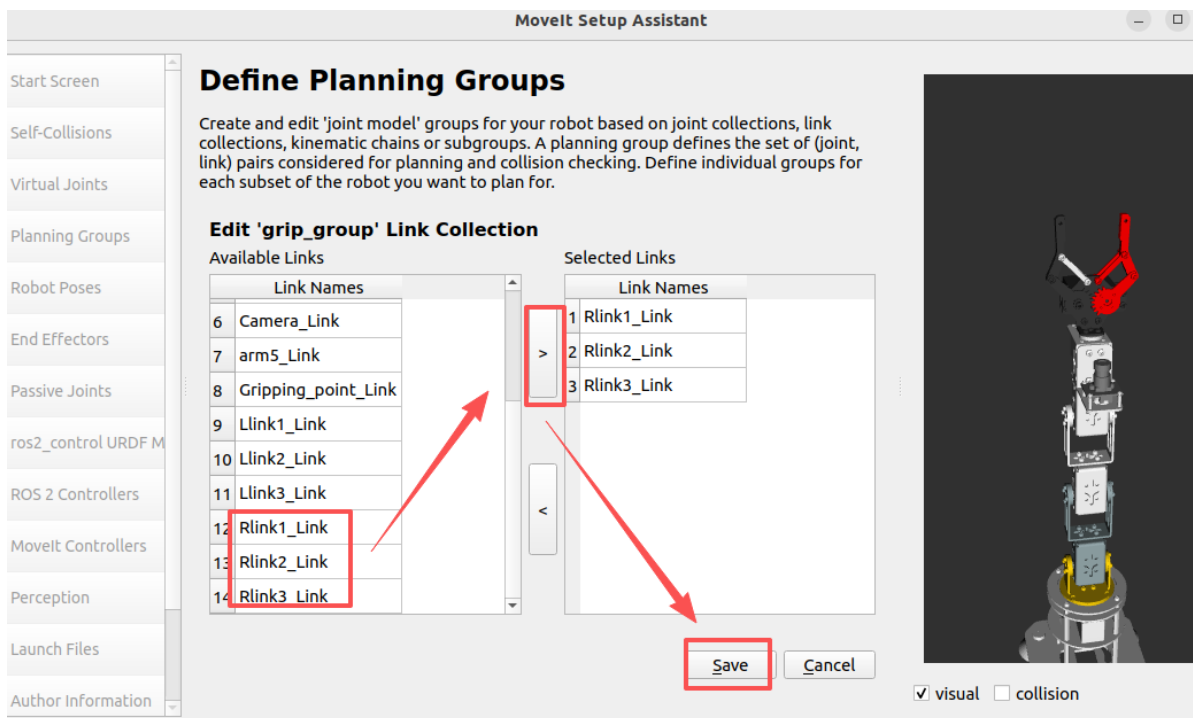
Plan gripper group: grip_group

- Group Name: grip_group
- Kinematic Solver: None
- Kin. Search Resolution: 0.005
- Kin. Search Timeout: 0.005



- Add Links: Select the gripper links

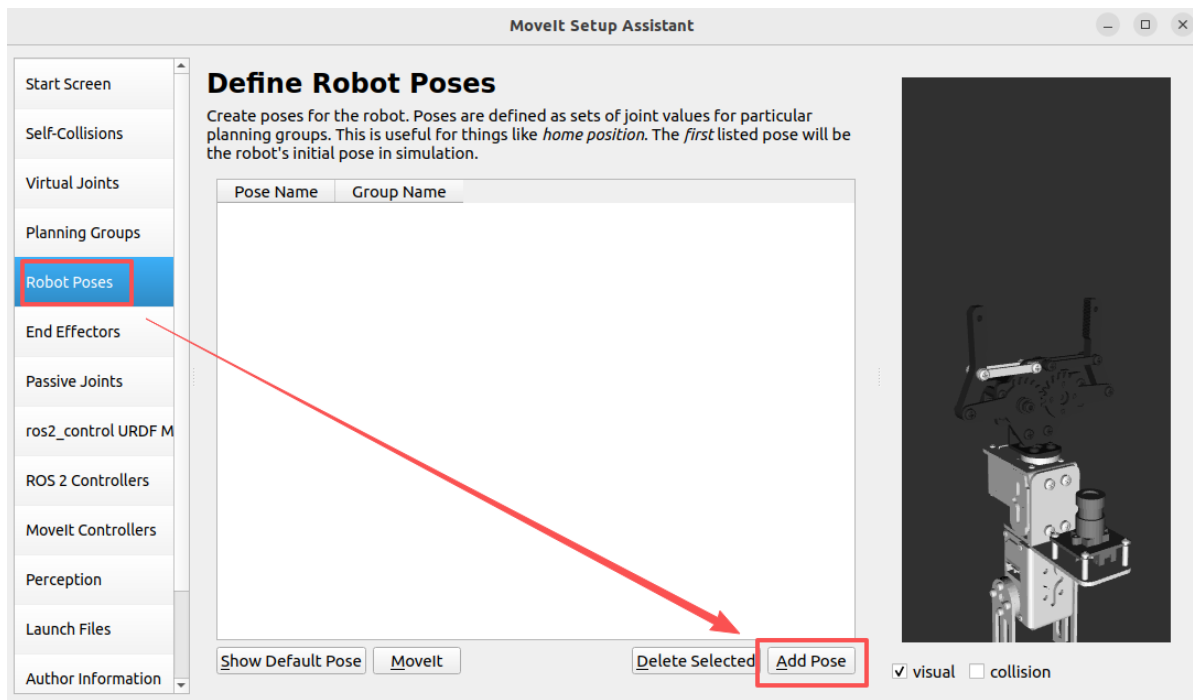
Links: rlink1, rlink2, rlink3



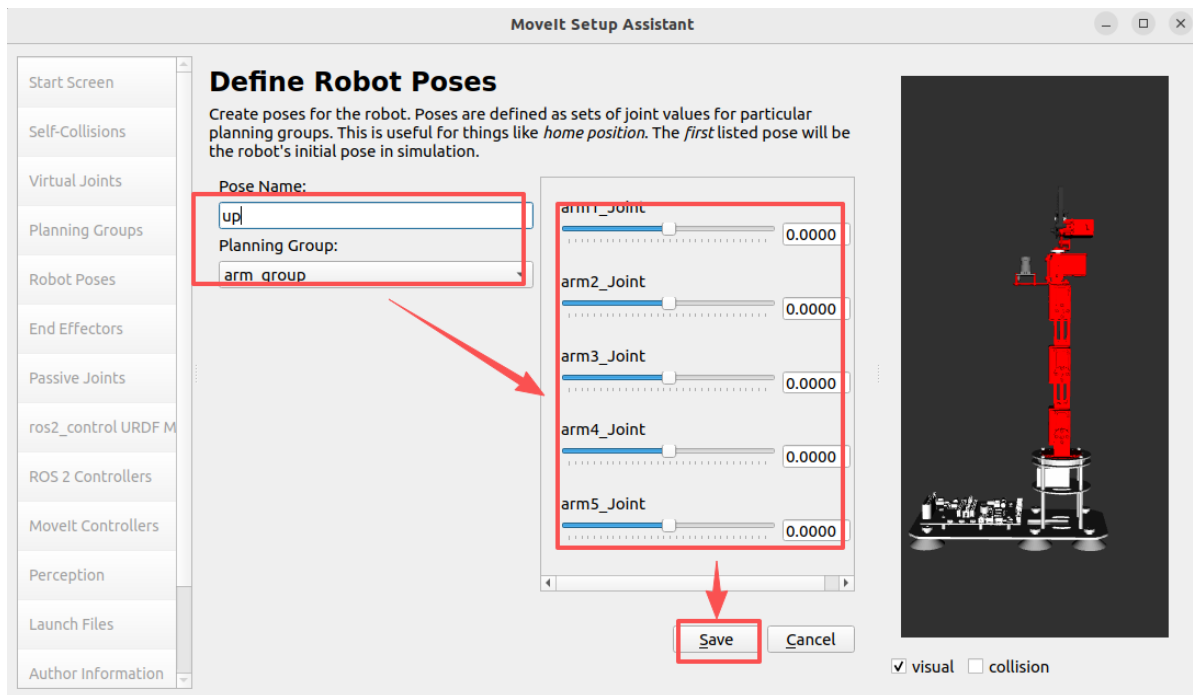
2.5. Robot Poses

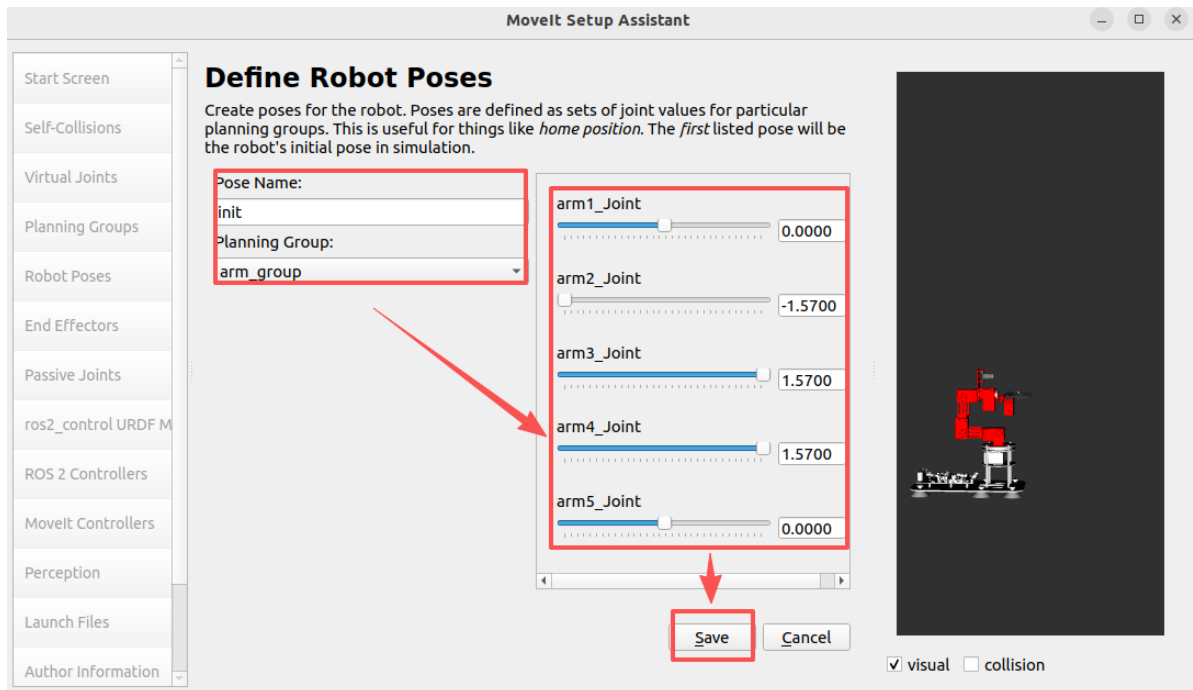
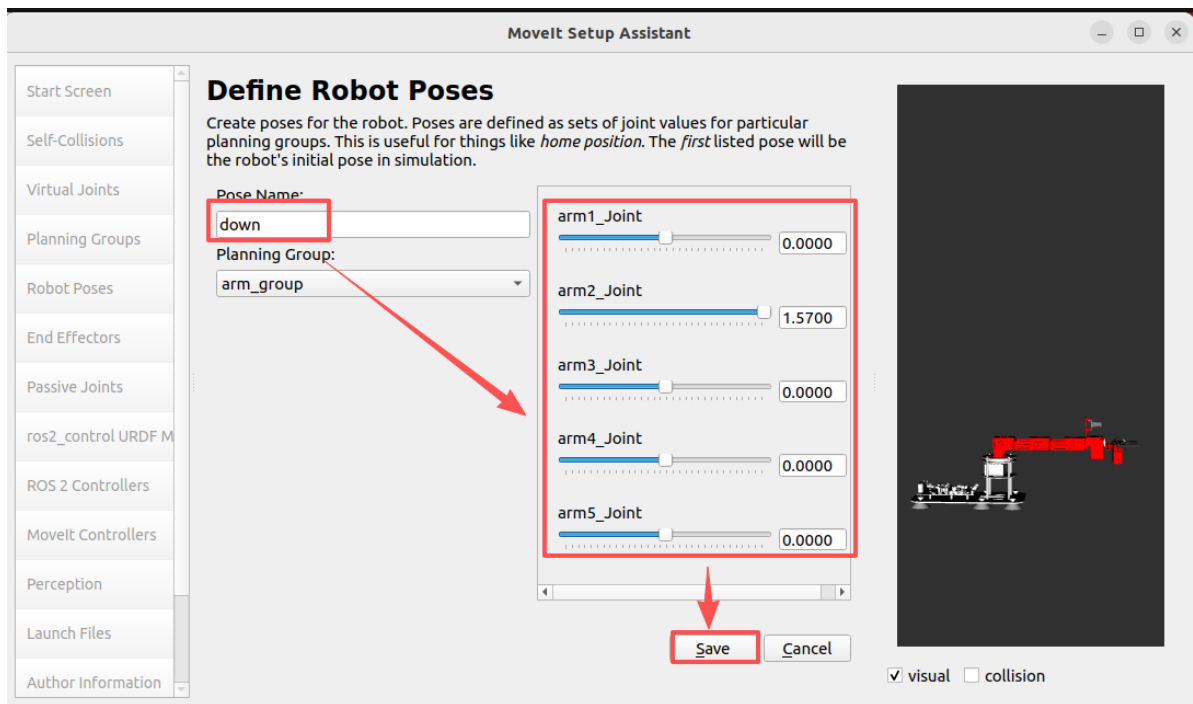
Robotic Arm Positions

Set three groups of robotic arm predefined poses to the robot configuration:



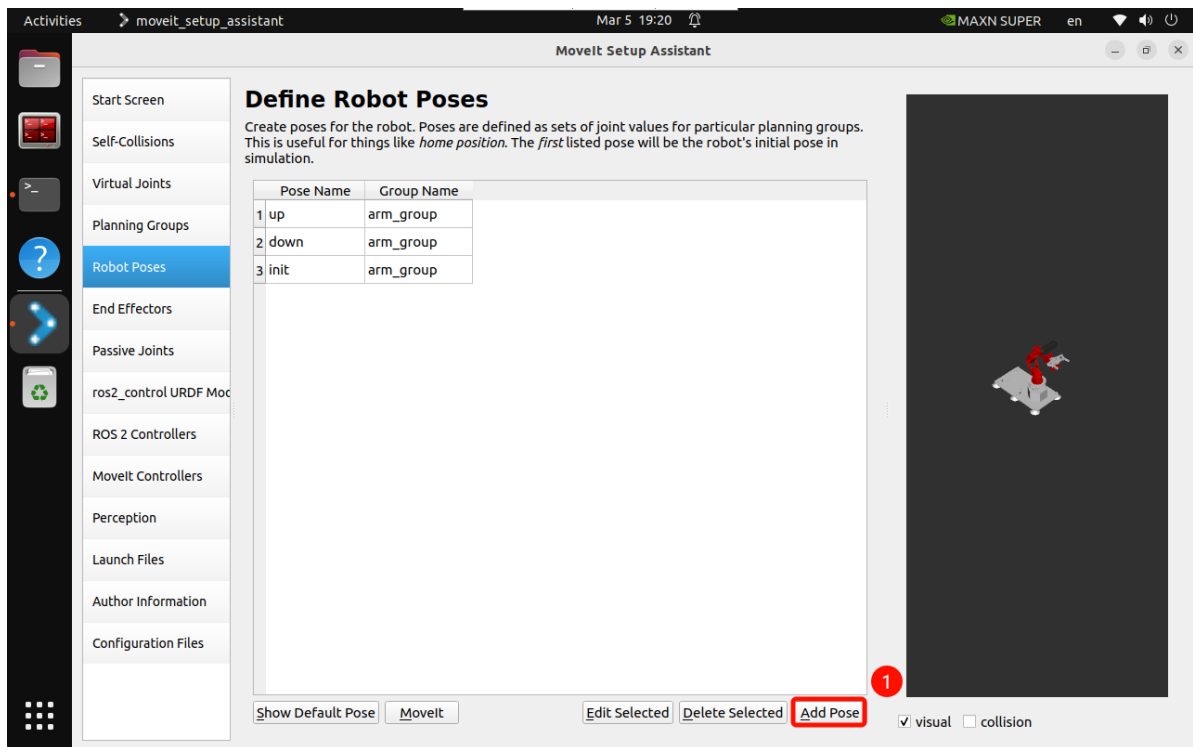
You can manually move the sliders to set each joint state: up, down, init



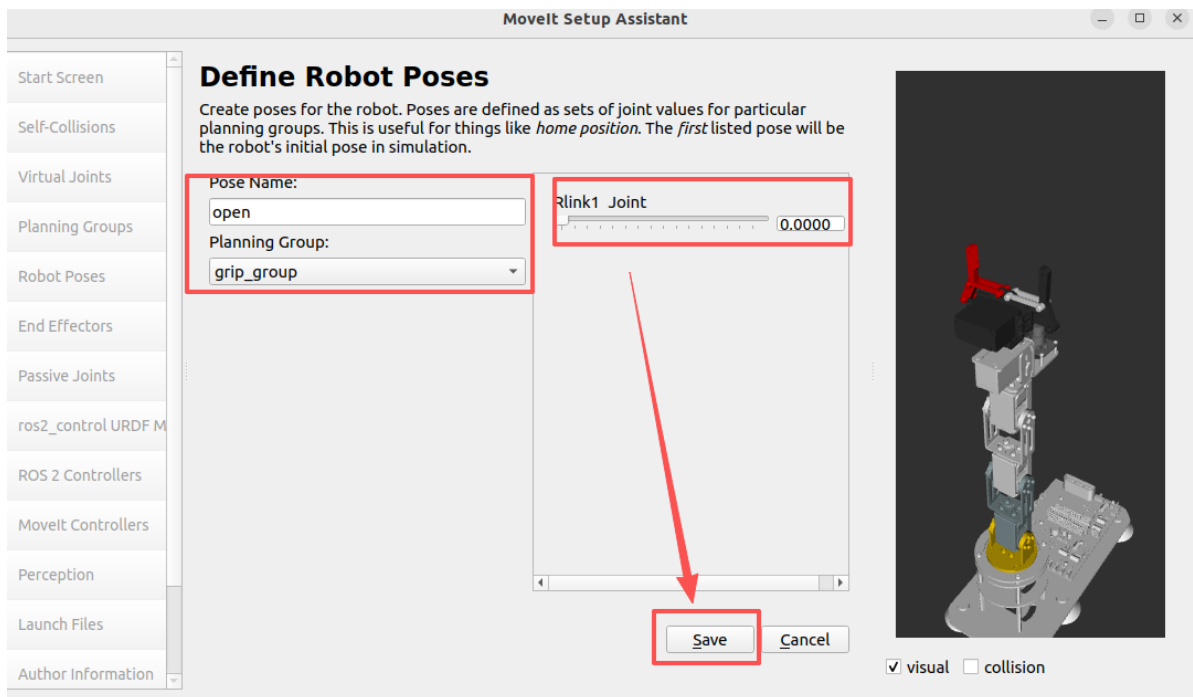


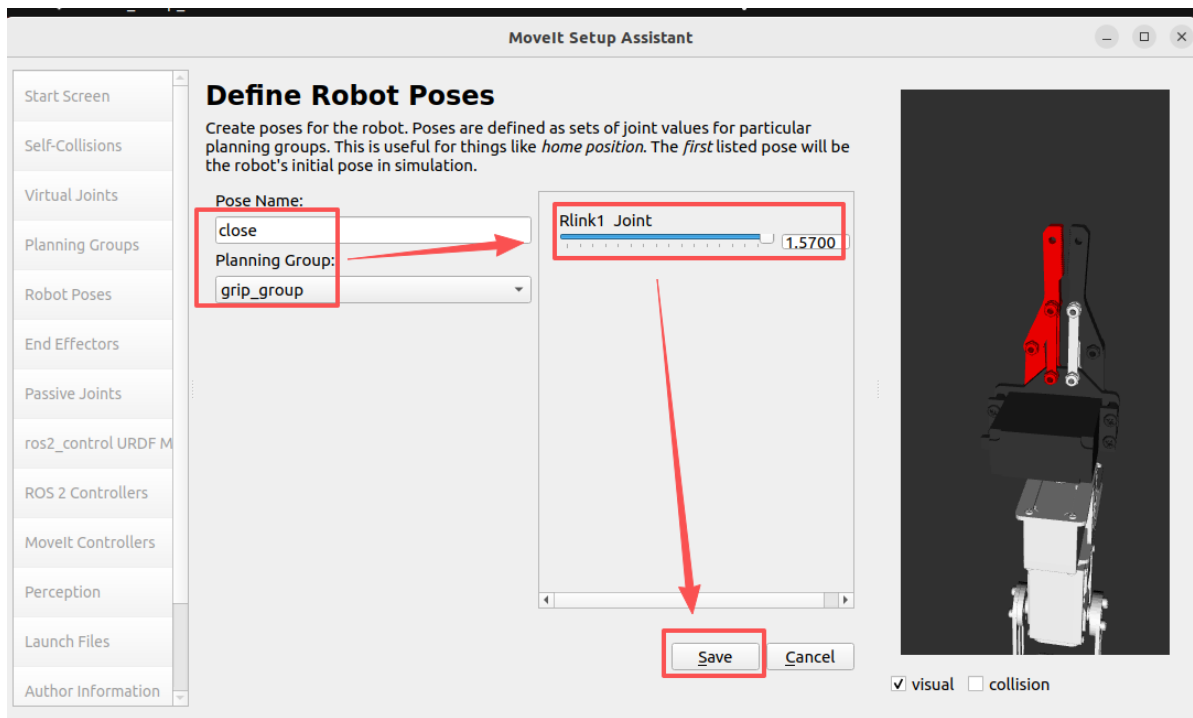
Gripper Positions

Set two groups of gripper predefined poses to the robot configuration:



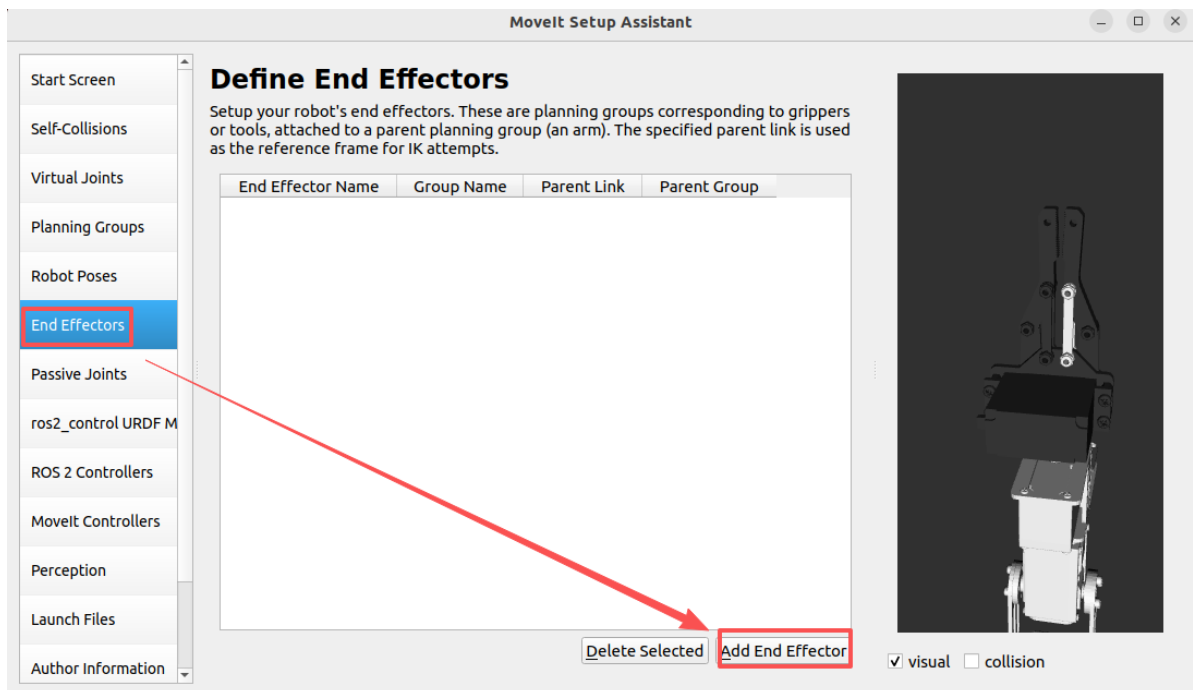
You can manually move the sliders to set gripper state: open, close





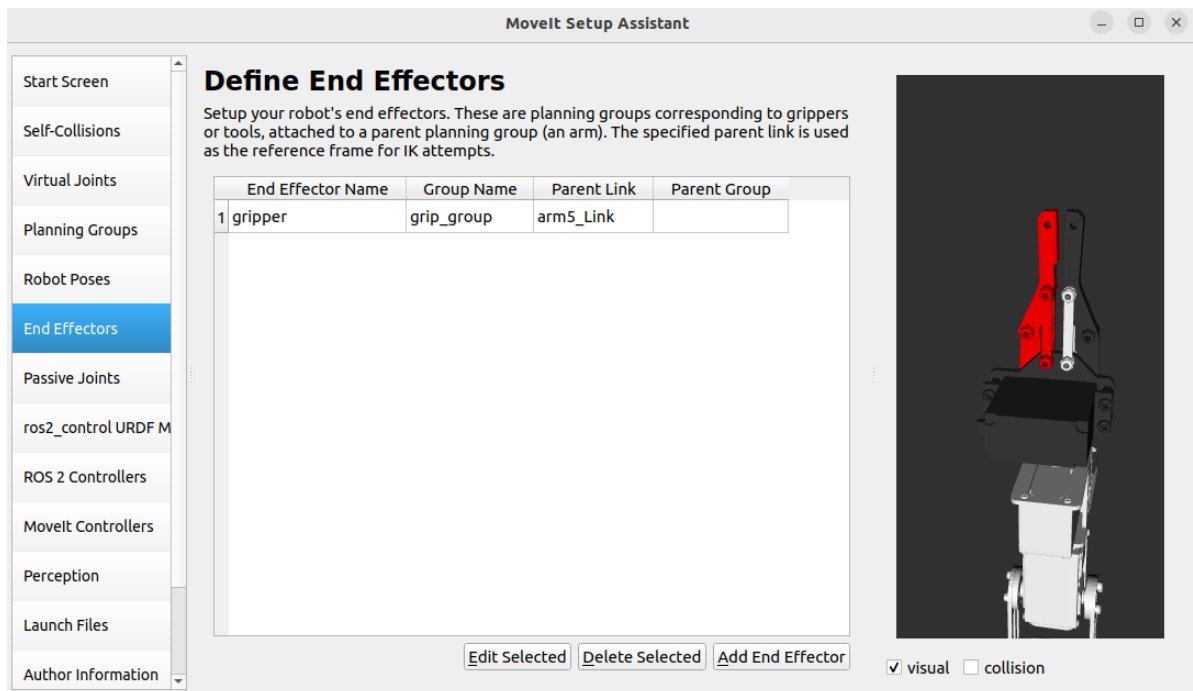
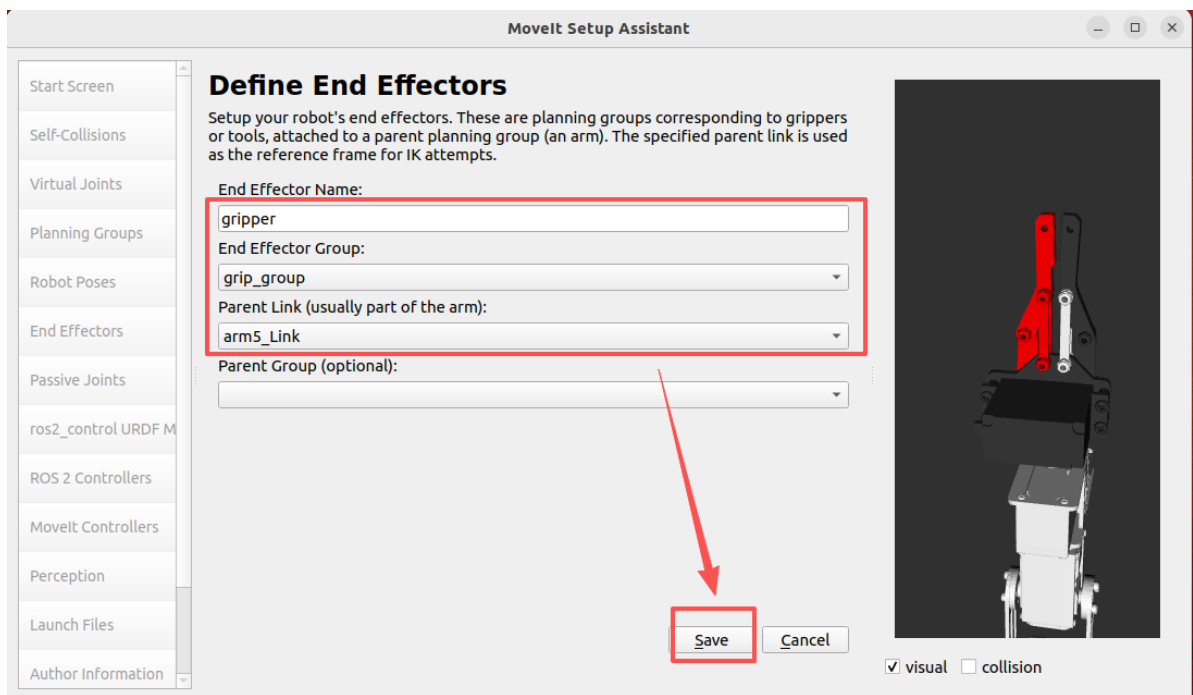
2.6. End Effector

Set the gripper group as the end effector:



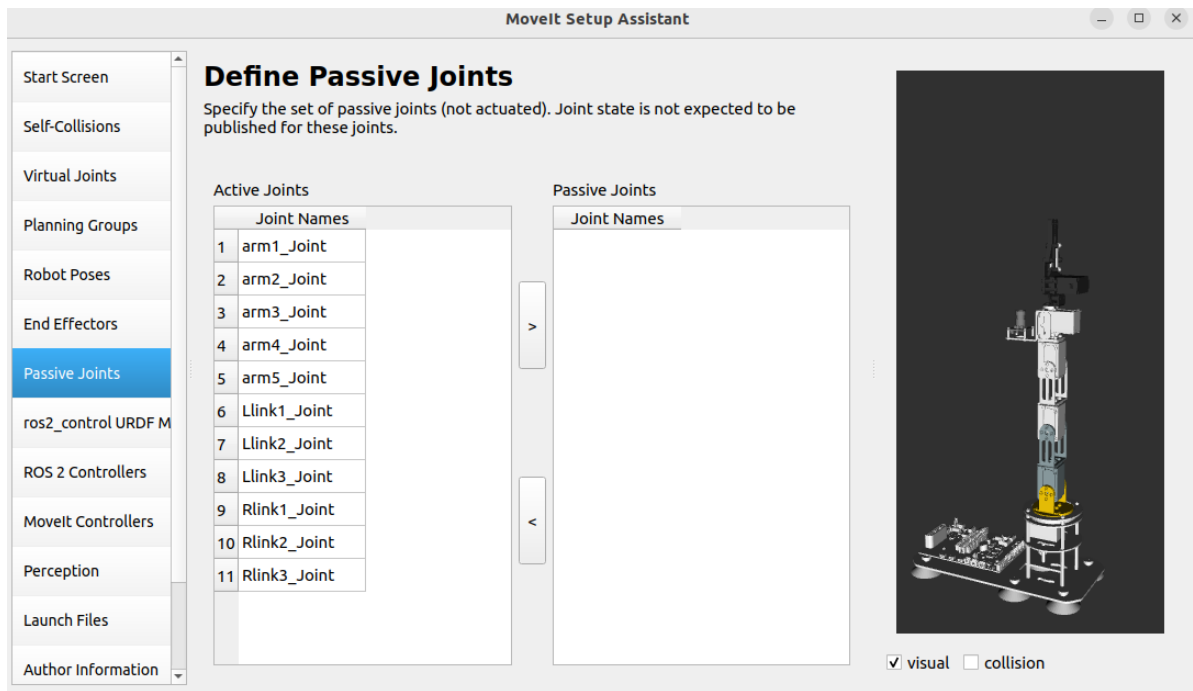
End effector: gripper

- End Effector Name: gripper
- End Effector Group: gripp_group
- Parent Link: Arm5_Link
- Parent Group: None



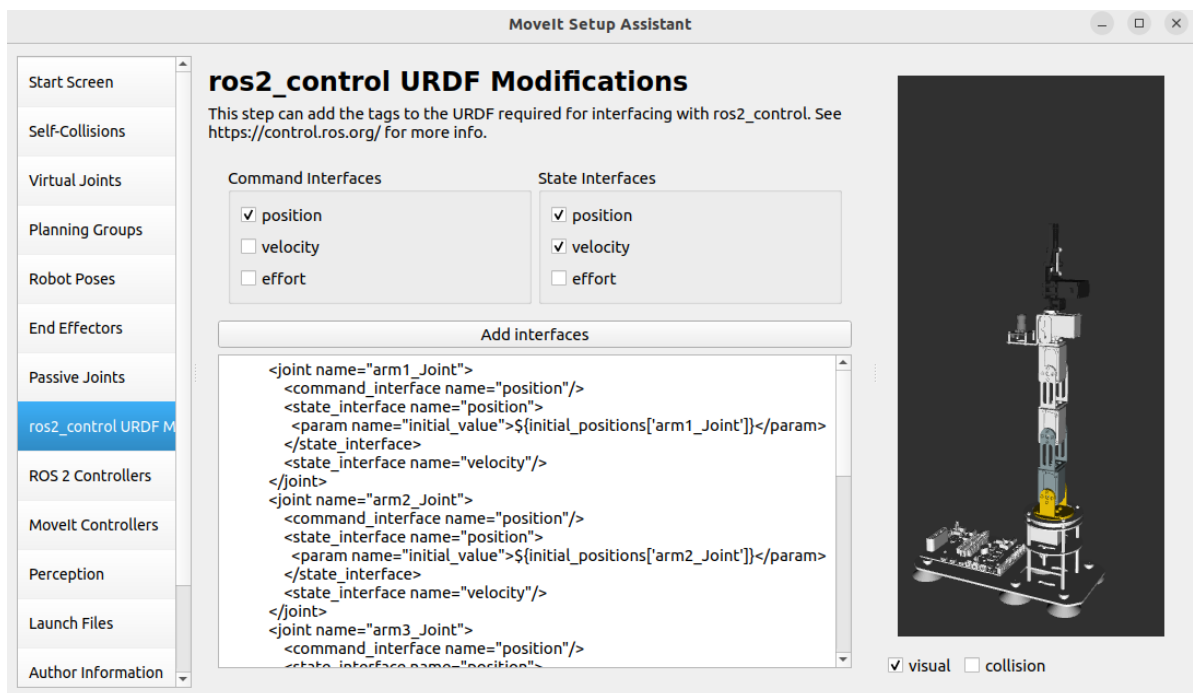
2.7. Passive Joints

Passive joints are non-driven joints that cannot be directly controlled: The robotic arm has no passive joints, so skip this step.



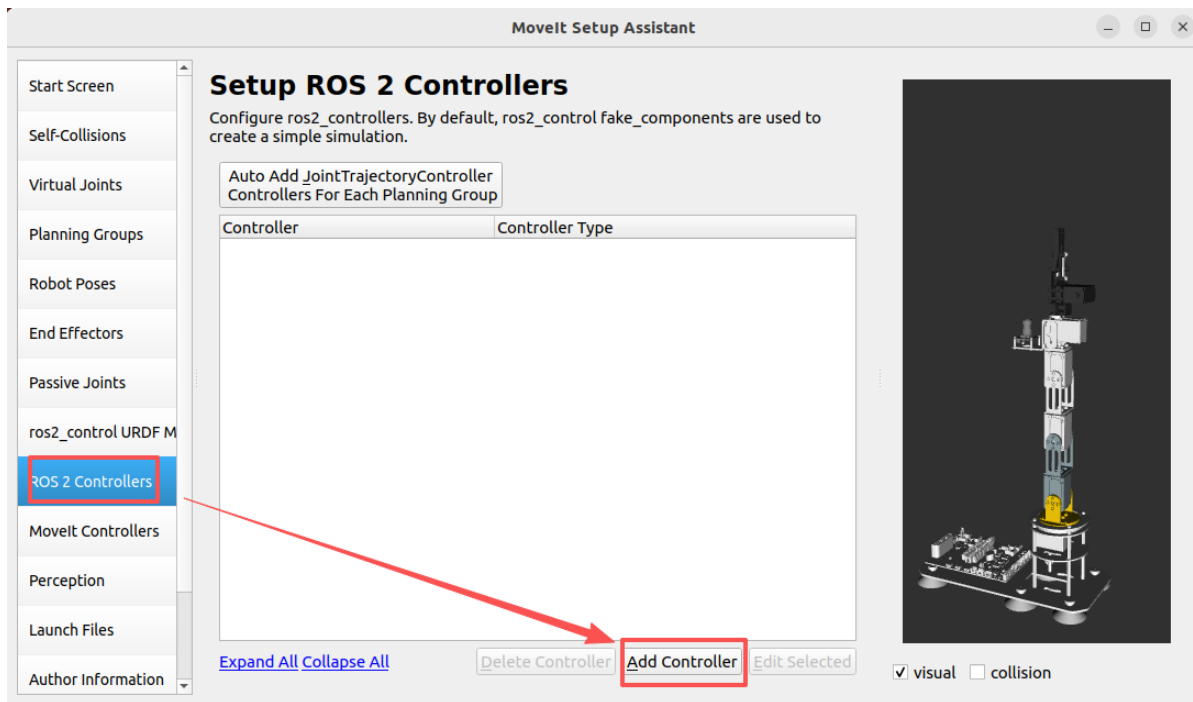
2.8. URDF Files

MoveIt Setup Assistant will automatically set command interfaces and state interfaces for each joint: Use the default generated information.



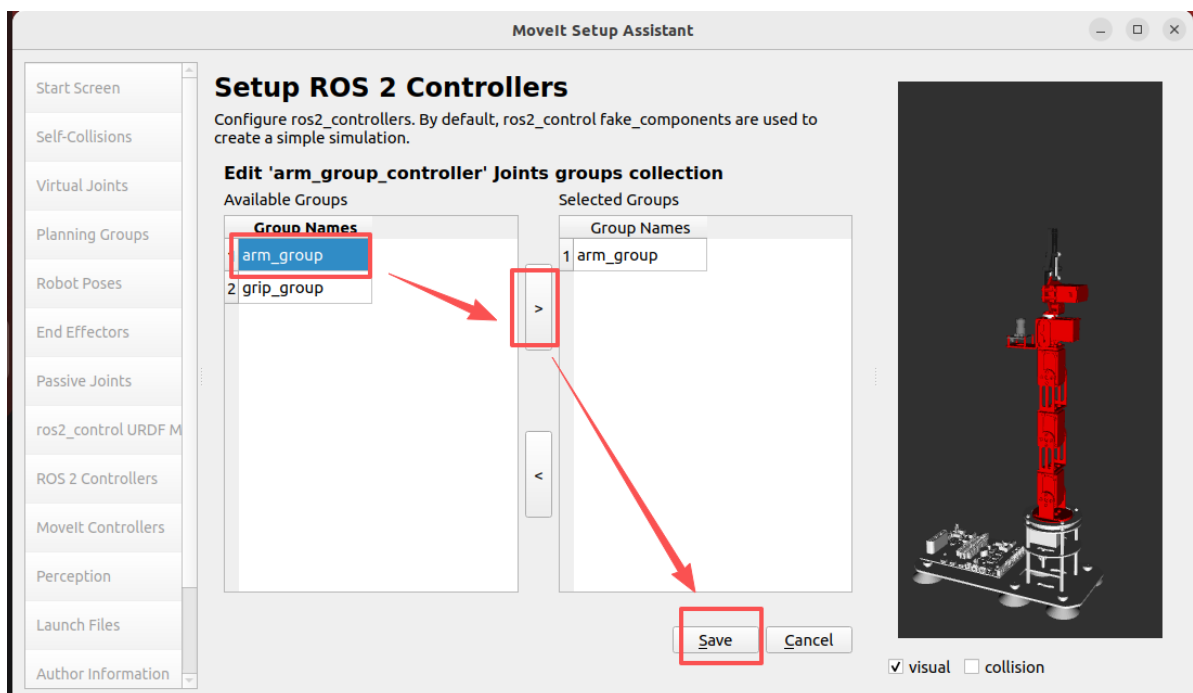
2.9. ROS2 Controllers

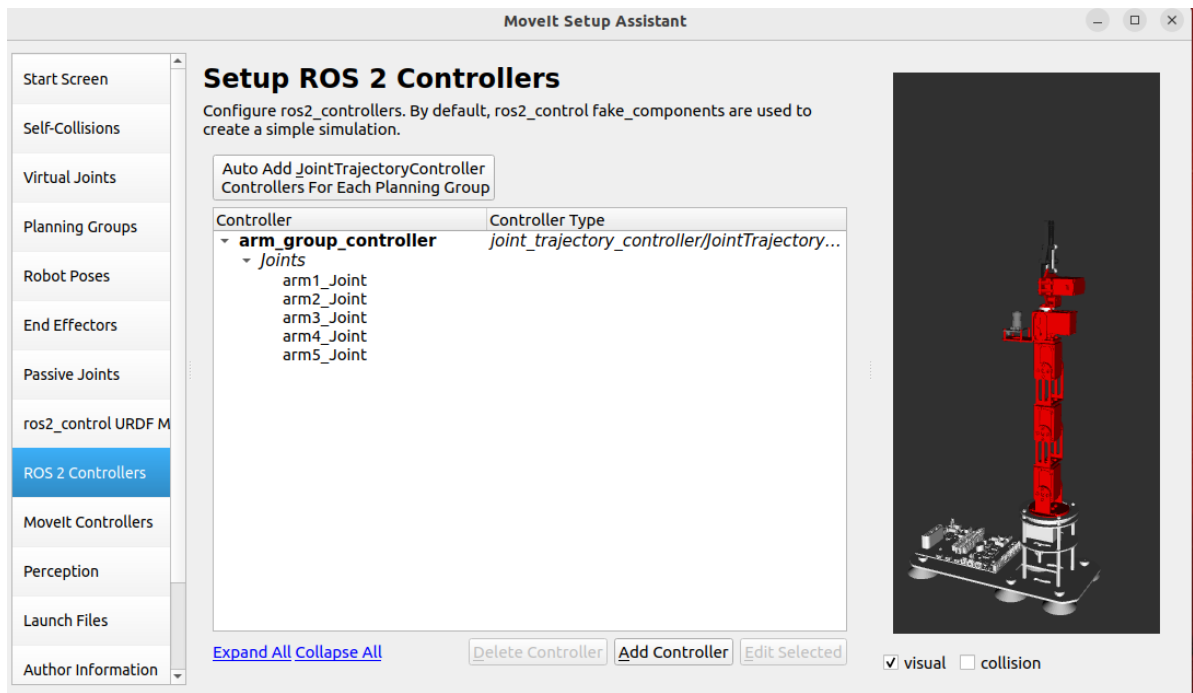
ROS2 Controllers is a framework for robot real-time control, which can be used to automatically generate simulation controllers to drive robot joints.



Add robotic arm controller:

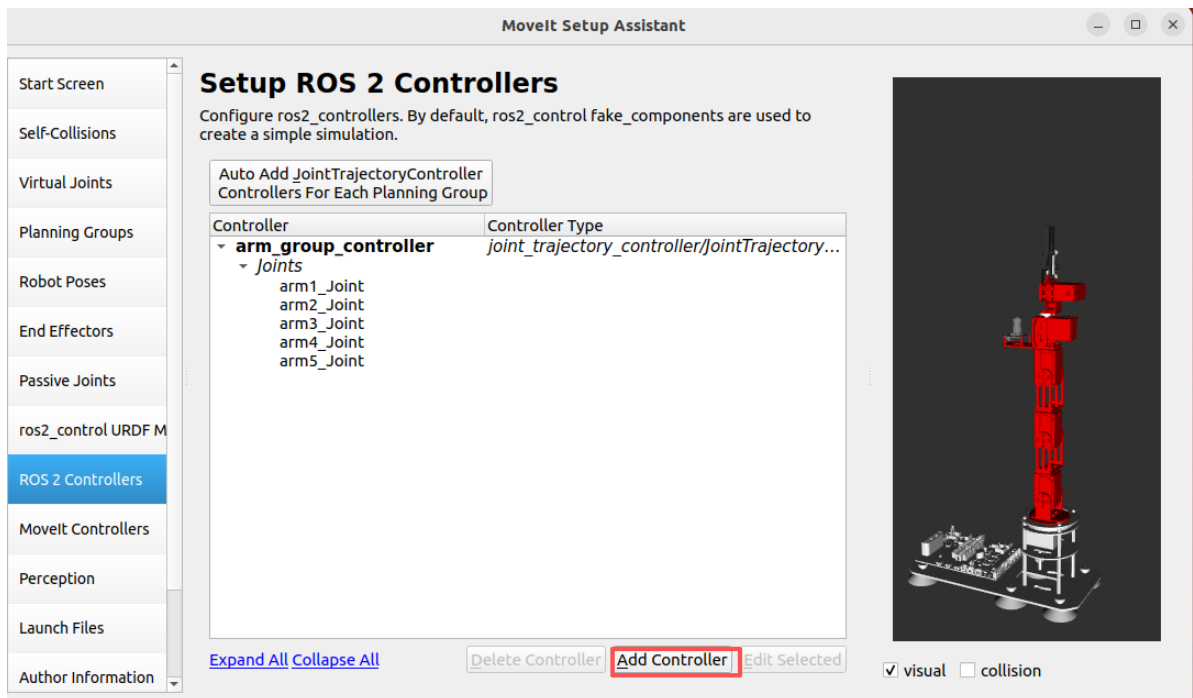
- Controller Name: arm_group_controller
- Controller Type: joint_trajectory_controller/JointTrajectoryController
- Add Planning Group Joints: arm_group

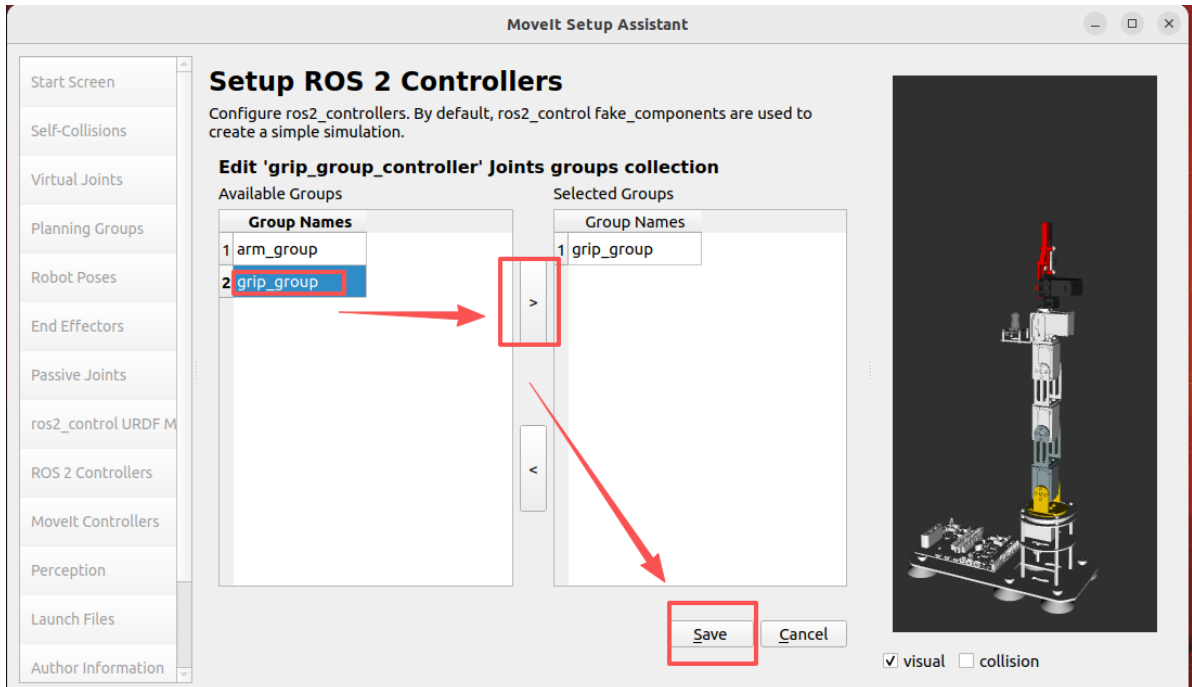
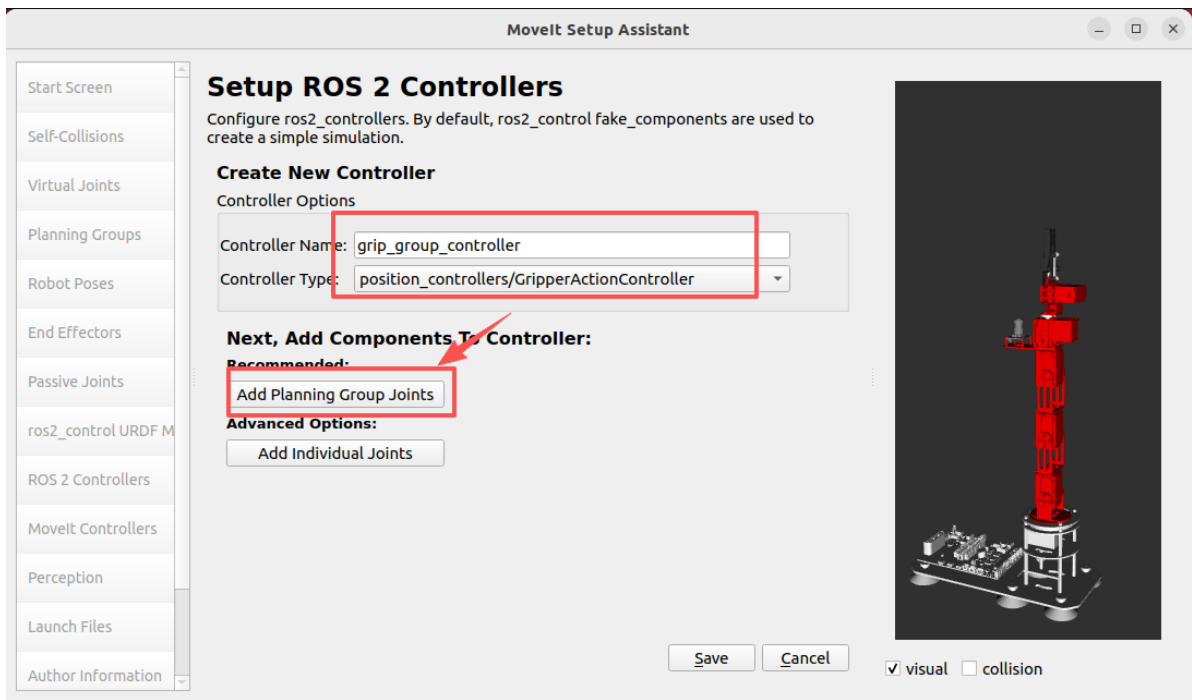


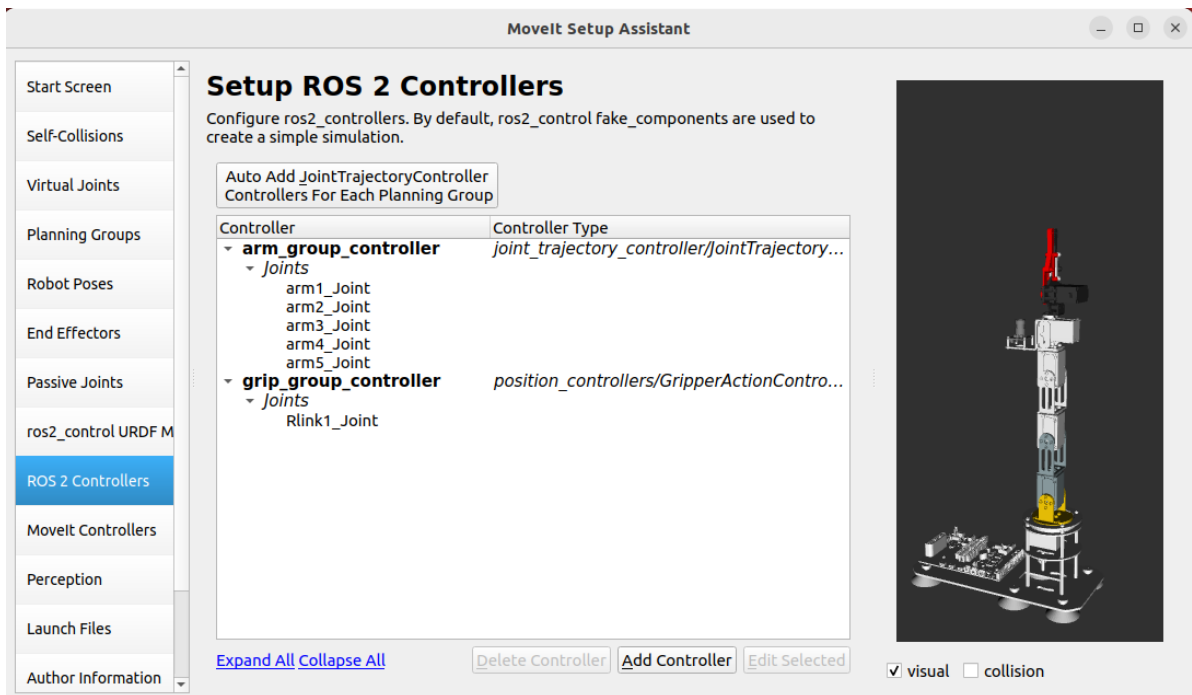


Add gripper controller:

- Controller Name: grip_group_controller
- Controller Type: position_controllers/GripperActionController
- Add Planning Group Joints: grip_group



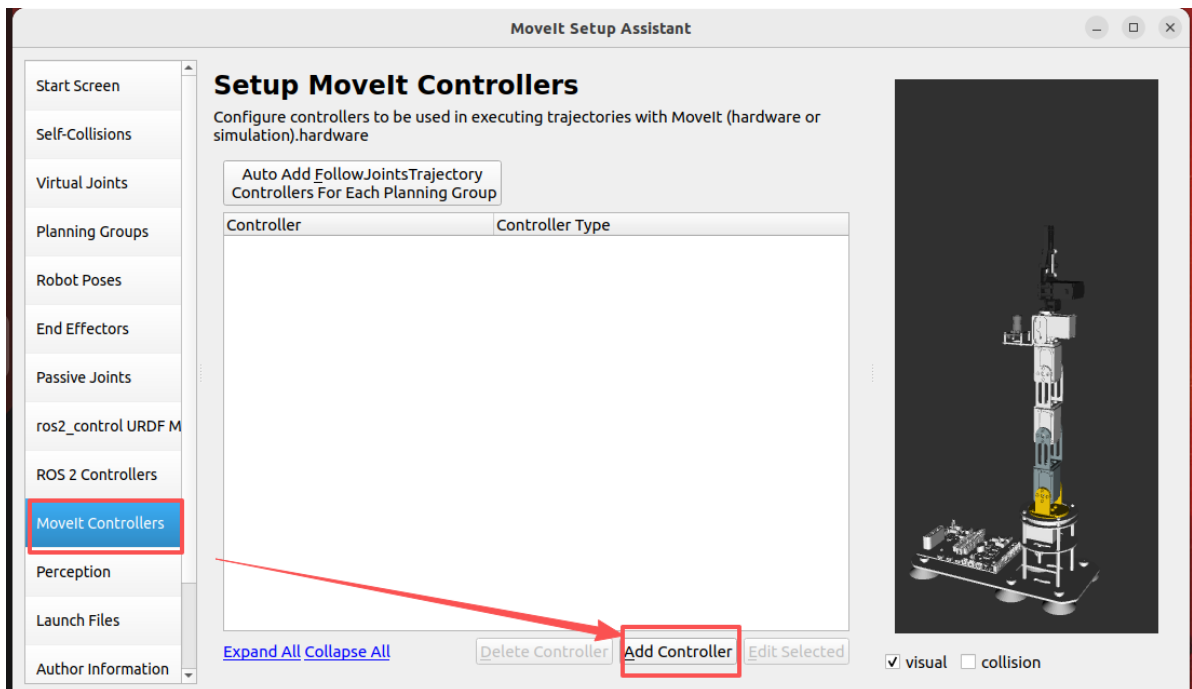




2.10. MoveIt Controllers

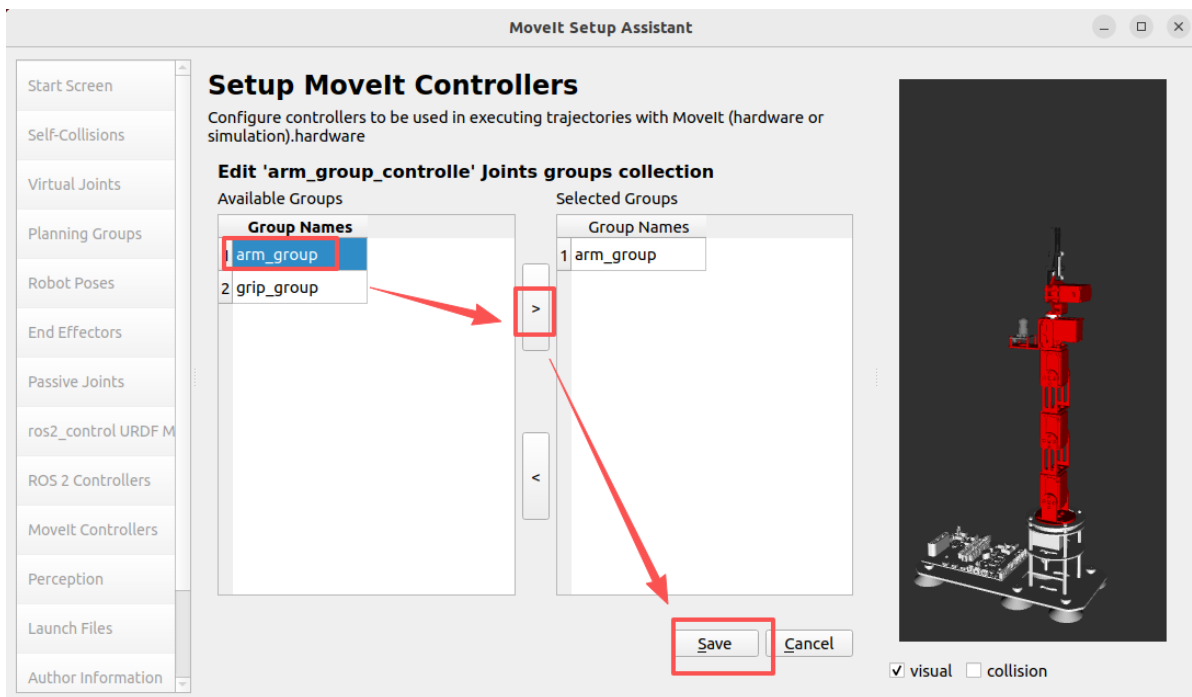
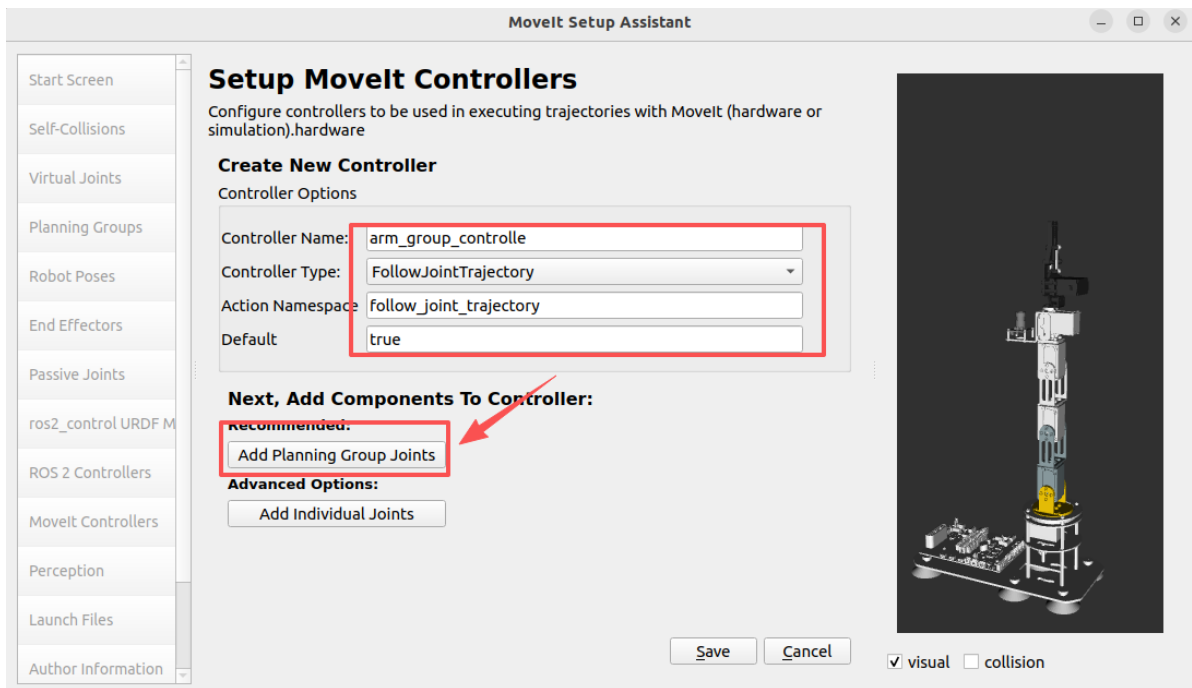
MoveIt requires trajectory controllers with the FollowJointTrajectoryAction interface to execute planned trajectories. This interface sends the generated trajectories to the robot's ROS2 controllers.

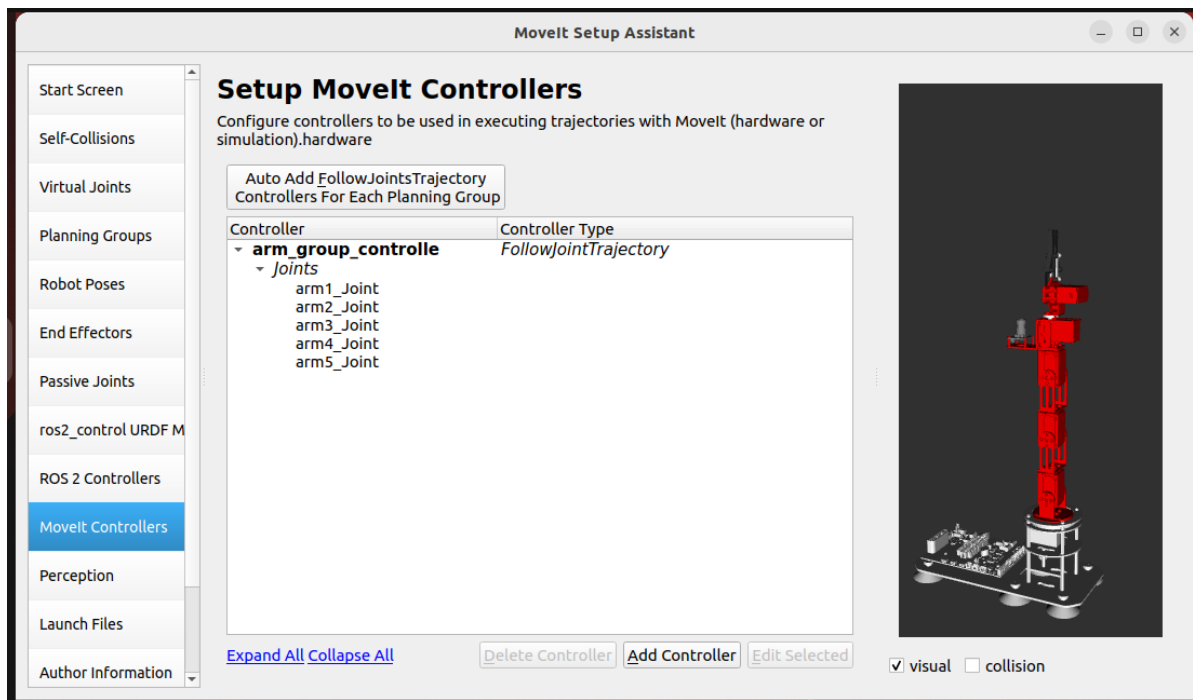
The added MoveIt Controllers need to ensure that the controller names match the names configured in ROS2 Controllers.



Add robotic arm controller:

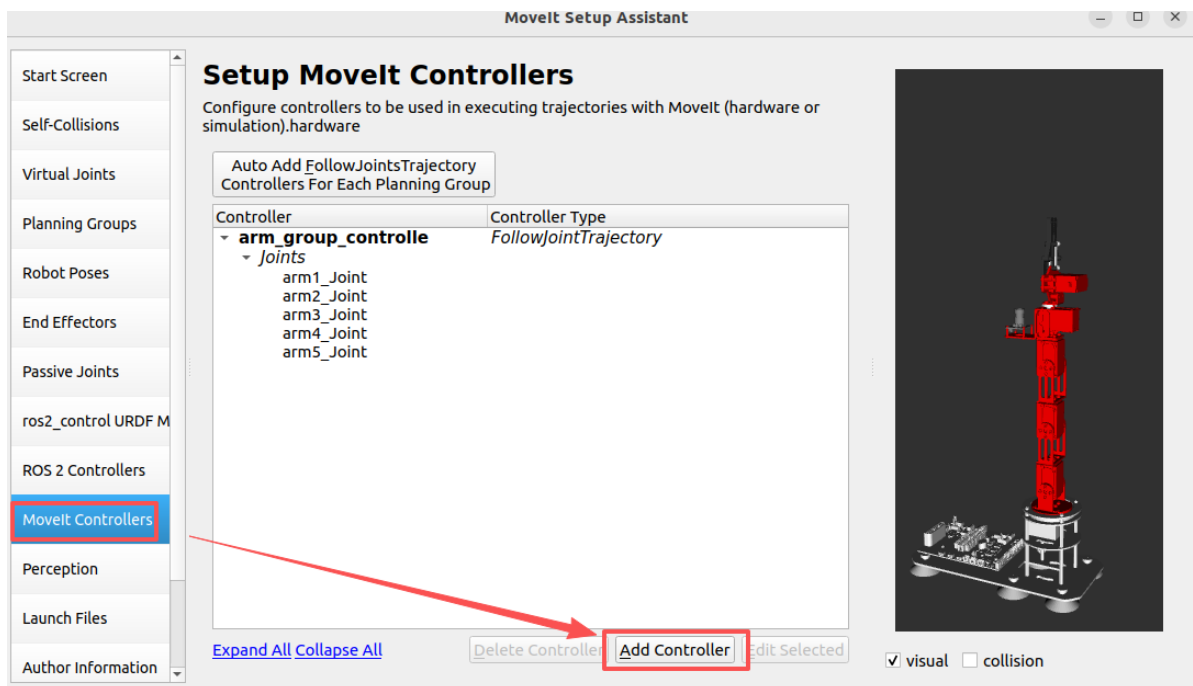
- Controller Name: arm_group_controller
- Controller Type: FollowJointTrajectory
- Add Planning Group Joints: arm_group





Add gripper controller:

- Controller Name: grip_group_controller
- Controller Type: Gripper Command
- Add Planning Group Joints: grip_group



Start Screen

Self-Collisions

Virtual Joints

Planning Groups

Robot Poses

End Effectors

Passive Joints

ros2_control URDF M

ROS 2 Controllers

MoveIt Controllers

Perception

Launch Files

Author Information

Setup MoveIt Controllers

Configure controllers to be used in executing trajectories with MoveIt (hardware or simulation).hardware

Create New Controller

Controller Options

Controller Name:

grip_group_controller

Controller Type:

GripperCommand

Action Namespace:

gripper_cmd

Default:

true

Next, Add Components To Controller:

Recommended:

Add Planning Group Joints

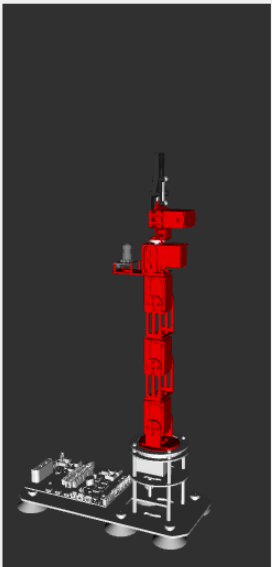
Advanced Options:

Add Individual Joints

Save

Cancel

☒ visual ☐ collision



Start Screen

Self-Collisions

Virtual Joints

Planning Groups

Robot Poses

End Effectors

Passive Joints

ros2_control URDF M

ROS 2 Controllers

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Setup MoveIt Controllers

Configure controllers to be used in executing trajectories with MoveIt (hardware or simulation).hardware

Edit 'grip_group_controller' Joints groups collection

Available Groups

	Group Names
1	arm_group
2	grip_group

Selected Groups

	Group Names
1	grip_group

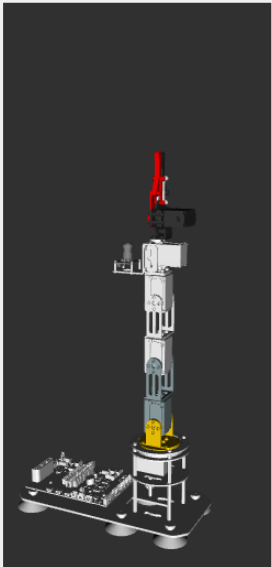
>

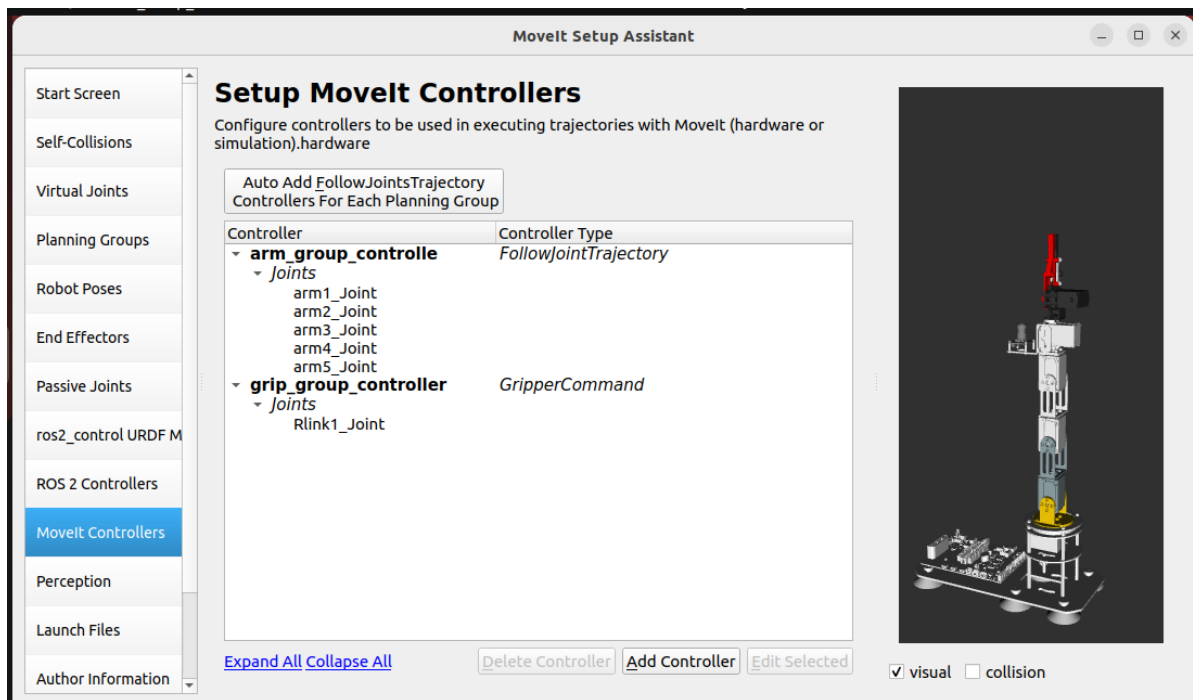
<

Save

Cancel

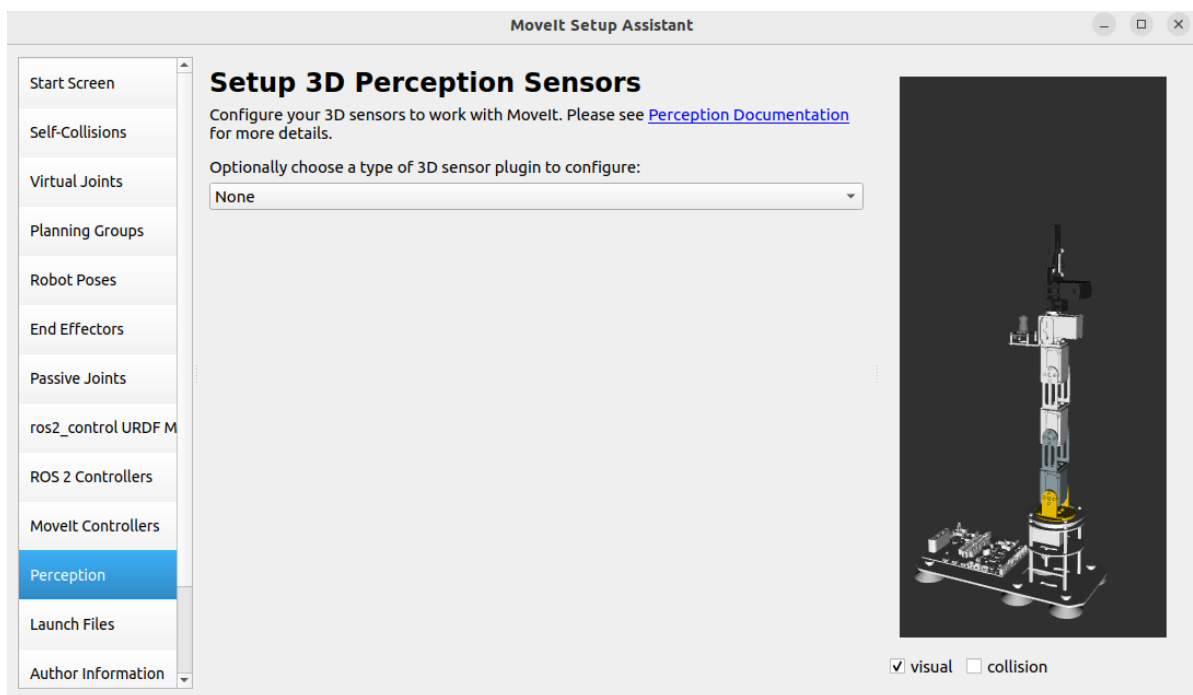
☒ visual ☐ collision





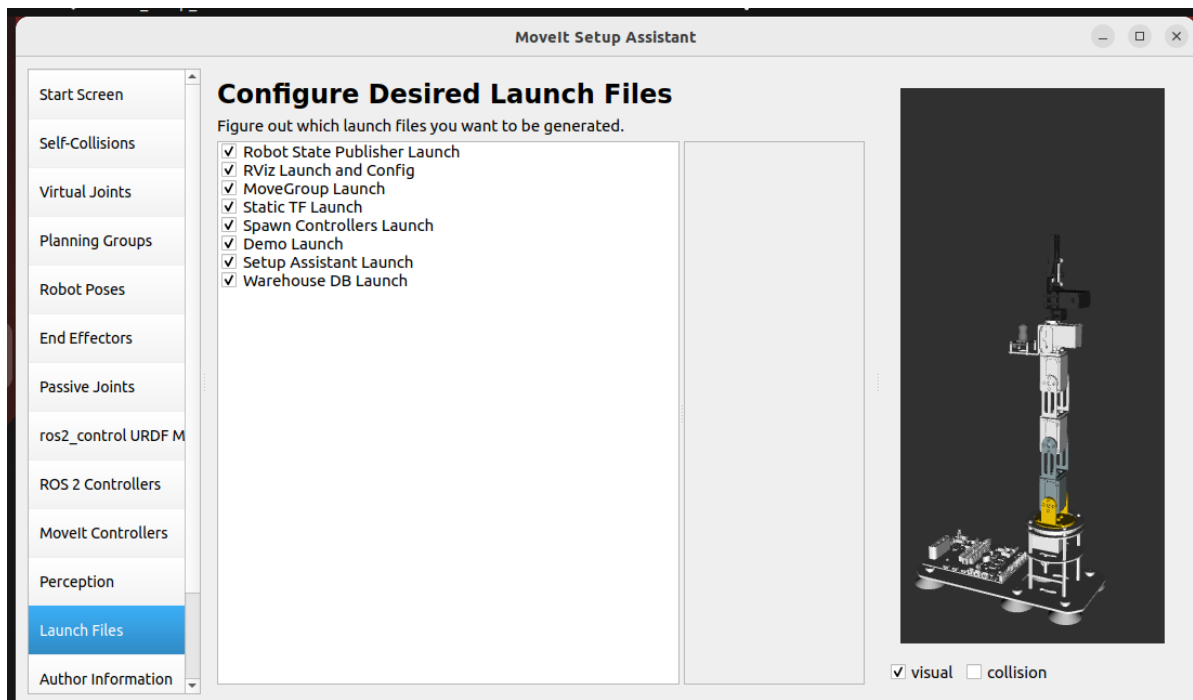
2.11. Sensors

Configure the settings for 3D sensors used by the robot: No configuration needed.



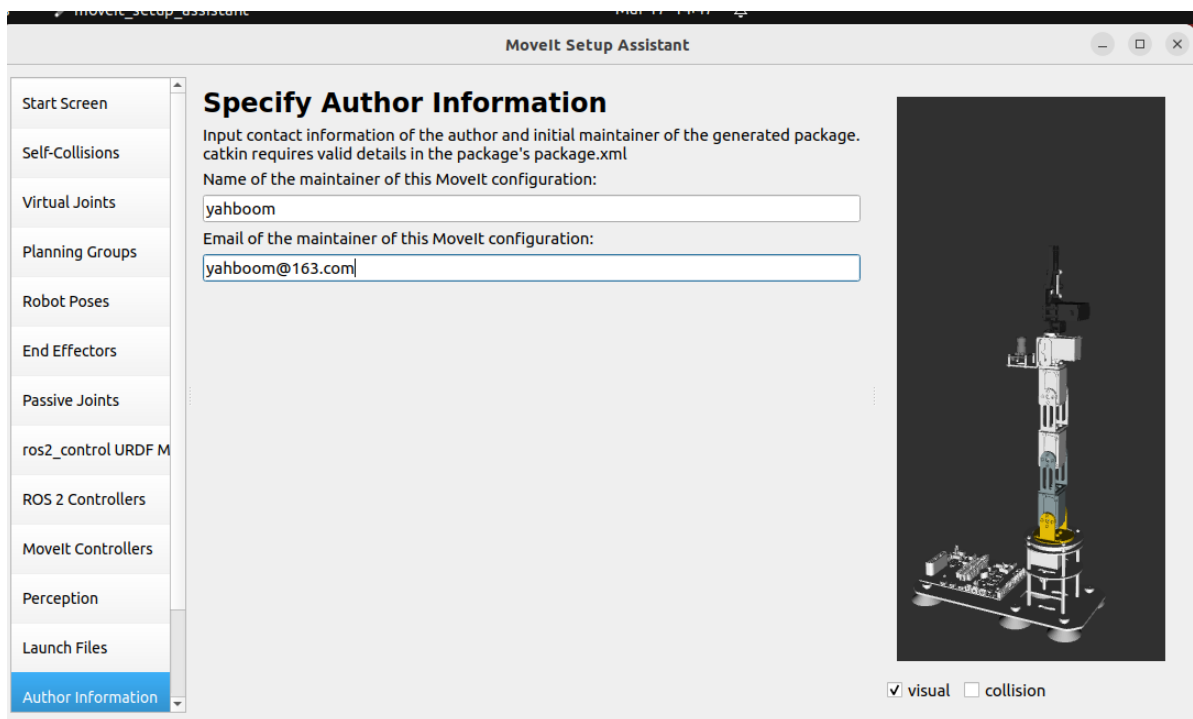
2.12. Launch Files

Configure auto-generated launch files: Use default options.



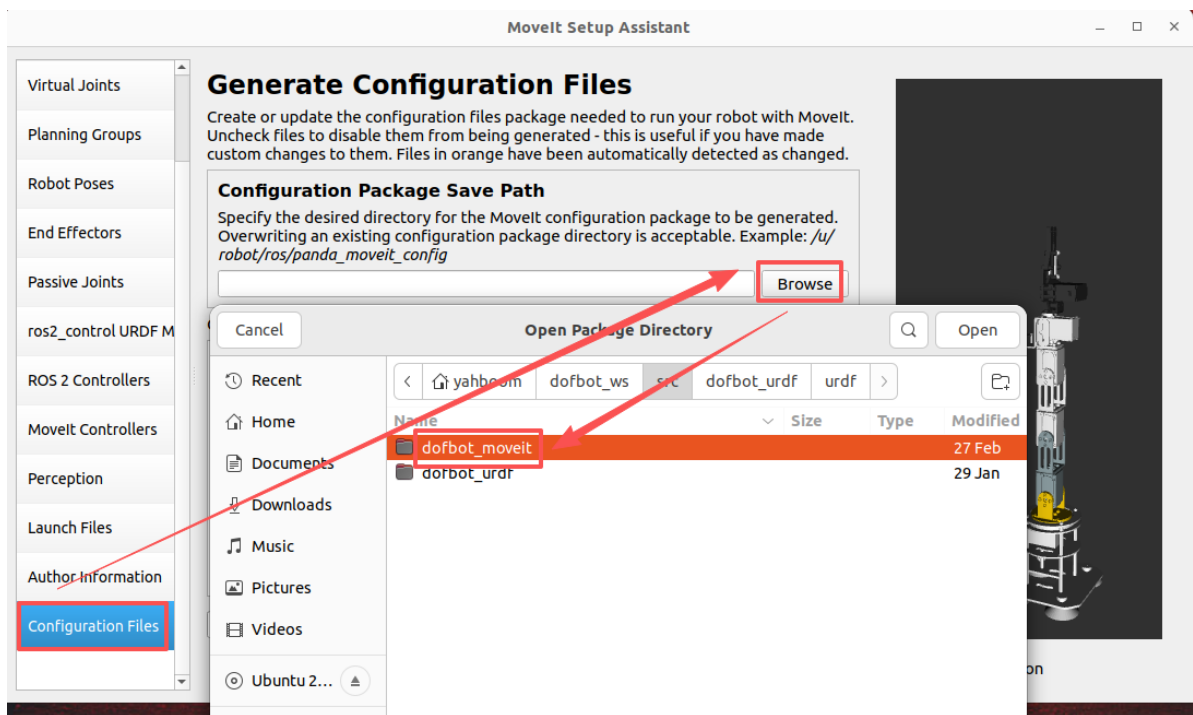
2.13. Author Information

Add author information for package generation:



2.14. Generate Configuration

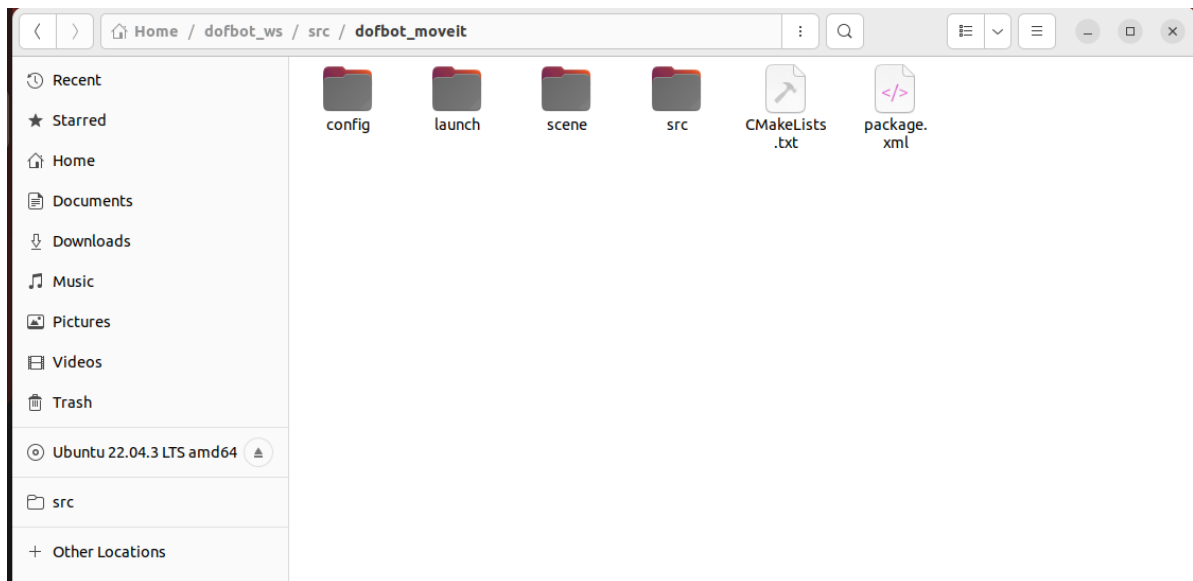
Generate configuration files to the specified folder:



Then click Generate Package.

3. Configuration Files

Enter the `dofbot_moveit` folder:



4. Configuration Verification

The configuration files generated by MoveIt Setup Assistant need simple modifications to remove startup warnings and model loading issues.

4.1. Modify Configuration Files

`dofbot_urdf.ros2_control.xacro`

Modify the `dofbot_urdf.ros2_control1.xacro` file: Modify the yaml loading function

```
dofbot_urdf.ros2_control.xacro
~/dofbot_ws/src/dofbot_moveit/config

1 <?xml version="1.0"?>
2 <robot xmlns:xacro="http://www.ros.org/wiki/xacro">
3   <xacro:macro name="dofbot_urdf_ros2_control" params="name initial_positions file">
4     <xacro:property name="initial_positions" value="${xacro.load_yaml(initial_positions_file)['initial_positions']}" />
5
6     <ros2_control name="${name}" type="system">
7       <hardware>
8         <!-- By default, set up controllers for simulation. This won't work on real hardware -->
9         <plugin>mock_components/GenericSystem</plugin>
10      </hardware>
11      <joint name="arm1_Joint">
12        <command_interface name="position"/>
13        <state_interface name="position">
14          <param name="initial_value">${initial_positions['arm1_Joint']}</param>
15        </state_interface>
16        <state_interface name="velocity"/>
17      </joint>
18      <joint name="arm2_Joint">
19        <command_interface name="position"/>
20        <state_interface name="position">
21          <param name="initial_value">${initial_positions['arm2_Joint']}</param>
22        </state_interface>
23        <state_interface name="velocity"/>
24      </joint>
25      <joint name="arm3_Joint">
26        <command_interface name="position"/>
27        <state_interface name="position">
28          <param name="initial_value">${initial_positions['arm3_Joint']}</param>
29        </state_interface>
30        <state_interface name="velocity"/>
31      </joint>
32      <joint name="arm4_Joint">
33        <command_interface name="position"/>
34        <state_interface name="position">
35          <param name="initial_value">${initial_positions['arm4_Joint']}</param>
36        </state_interface>
37        <state_interface name="velocity"/>
38      </joint>
39      <joint name="arm5_Joint">
```

Bracket match found on line: 4

XML Tab Width: 8 Ln 4, Col 119 INS

joint_limits.yaml

```
joint_limits.yaml
~/dofbot_ws/src/dofbot_moveit/config

6 default_acceleration_scaling_factor: 0.1
7
8 # Specific joint properties can be changed with the keys [max_position, min_position, max_velocity, max_acceleration]
9 # Joint limits can be turned off with [has_velocity_limits, has_acceleration_limits]
10 joint_limits:
11   Llink1_Joint:
12     has_velocity_limits: true
13     max_velocity: 1000.0
14     has_acceleration_limits: false
15     max_acceleration: 0.0
16   Llink2_Joint:
17     has_velocity_limits: false
18     max_velocity: 1.0
19     has_acceleration_limits: false
20     max_acceleration: 0.0
21   Llink3_Joint:
22     has_velocity_limits: false
23     max_velocity: 1.0
24     has_acceleration_limits: false
25     max_acceleration: 0.0
26   Rlink1_Joint:
27     has_velocity_limits: true
28     max_velocity: 1000.0
29     has_acceleration_limits: false
30     max_acceleration: 0.0
31   Rlink2_Joint:
32     has_velocity_limits: false
33     max_velocity: 1.0
34     has_acceleration_limits: false
35     max_acceleration: 0.0
36   Rlink3_Joint:
37     has_velocity_limits: false
38     max_velocity: 1.0
39     has_acceleration_limits: false
40     max_acceleration: 0.0
41   arm1_Joint:
42     has_velocity_limits: true
43     max_velocity: 1000.0
44     has_acceleration_limits: false
```

YAML Tab Width: 8 Ln 27, Col 30 INS

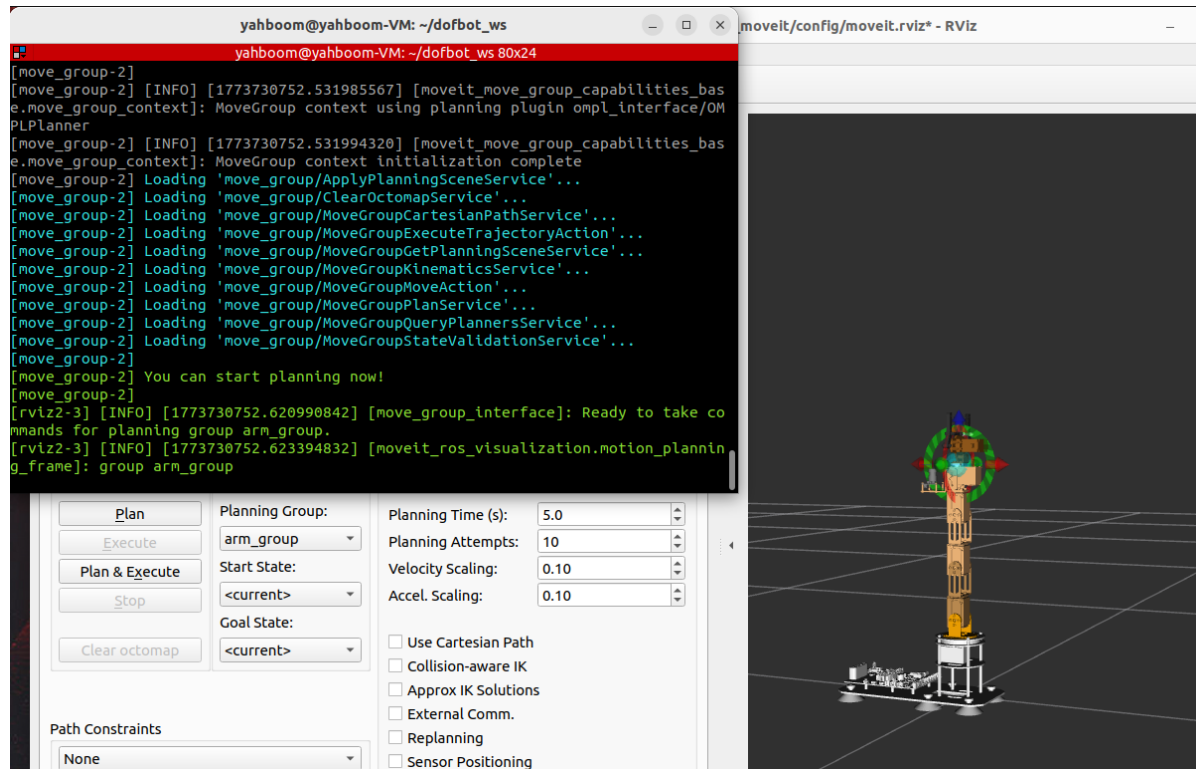
4.2. Compile Package

```
cd ~/dofbot_ws
colcon build --packages-select dofbot_moveit
source install/setup.bash
```

4.3. Launch MoveIt

```
ros2 launch dofbot_pro_moveit demo.launch.py
```

The simulation startup is relatively slow. Wait until the terminal shows `You can start planning now!` or the robotic arm shows trajectory balls (trajectory balls are new spheres added to the robotic arm), indicating loading is complete.



References

MoveIt2 Humble: <https://moveit.picknik.ai/humble/index.html>

MoveIt2: <https://moveit.picknik.ai/main/index.html>